



MSCC Operator's Guide — Linux (Ubuntu / Raspberry Pi)

Multus SDR Control Console for Proficio, Geminus and Ultimus transceivers

Instructions based on MSCC Avalonia (mscc 1.0.50 · msc-ui 0.6.72) · **Release Version 0**

This edition covers Linux local operation on Ubuntu Desktop and Raspberry Pi, firmware updates (Appendix A), and Appendices B–G (keyer, installing and upgrading, settings and calibration files, troubleshooting, WSJT-X, and Raspberry Pi setup differences).

Remote operation has its own guide.

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□ WELCOME

This guide covers installing and using MSCC, the Multus SDR Control Console, on a **Linux PC running Ubuntu Desktop (64-bit)** or a **Raspberry Pi 4 or 5 running 64-bit Raspberry Pi OS**, with the radio plugged into that same computer (“local operation”). MSCC is the control program for Multus SDR, LLC transceivers: Proficio, Geminus and Ultimus, in both Legacy and MKII versions. It works only with Multus SDR equipment.

MSCC comes in two parts:

- **The servers.** These small programs talk to the radio over USB and do the receive and transmit signal processing. They run on the computer the radio is plugged into. Their names are `sdrcore-recv`, `sdrcore-trans` and `ms-sdr`; you will see these names in some messages.
- **The client.** This is the window you operate from. In the menu it is called **MSCC UI**; its window title reads **MSCC Avalonia**. These instructions are based on MSCC Avalonia.

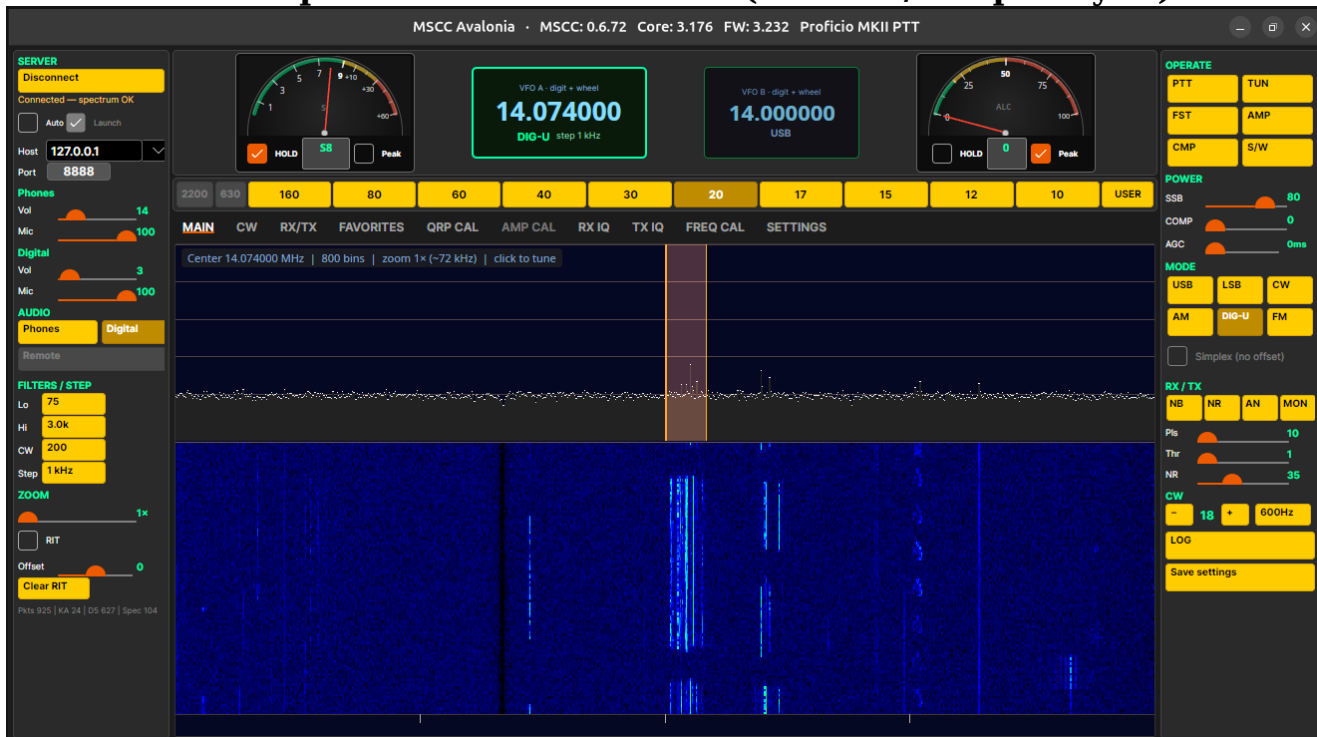
A setup program, **MSCC Init**, tells the servers which radio, serial port, speaker and microphone to use. You run it once after installing, and again whenever you change your audio devices.

MSCC UI looks and works the same on Ubuntu and on the Raspberry Pi. Only installing differs; the Pi differences are in Appendix G. Pictures in this guide were taken on an Ubuntu PC.

The client and servers talk over your network connection, even on the same computer. Operating from another computer (“remote”) has its own guide.



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MSCC UI connected to the radio, MAIN tab showing



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□ 1. Overview (READ THIS FIRST!)

Please read this section before you install anything.

1. **Firmware first.** The radio's firmware should match this version of MSCC. The window title shows the firmware version (FW) after you connect. Firmware updates are covered in Appendix A; the firmware files come in the optional `mscc-firmware` package.
2. **Your setup.** This guide is for *Linux local*: the radio plugs into an Ubuntu PC or a Raspberry Pi, and MSCC UI on that same computer connects to the servers at 127.0.0.1. For remote operation, see the separate guide.
3. **Set up once with MSCC Init.** After installing, run MSCC Init (2.2). MSCC UI warns you if it hasn't been done.
4. **Calibrate once.** A new radio or new installation should get frequency calibration (FREQ CAL), power calibration (QRP CAL) and transmit image balance (TX IQ). Factory values are loaded automatically for your radio model, so this is a touch-up, not a must-do on day one. Press **Save settings** when you finish calibrating (saves TX IQ, QRP and AMP calibration on this computer for this radio).
5. **Version numbers in this guide are examples.** File names such as `mscc_1.0.xx_amd64.deb` stand for whatever version you were given. Always use the newest files you have.
6. **Bug reports and feature requests:** post them on our forum, <https://multusdr.groups.io/g/main/topics>



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□ 2. Linux Local Operation (Instructions based on MSCC Avalonia)

2.1 Installing and starting for the first time

What you need: an Ubuntu Desktop 64-bit PC (or a Raspberry Pi 4/5, Appendix G); the radio with its USB cable plugged in; a speaker or headphones and a microphone on that computer; the MSCC install files for your computer in one folder; a mouse with a scroll wheel.

The **Ubuntu kit** contains `mscc-portaudio_..._amd64.deb`, `install-mscc.sh`, `mscc_..._amd64.deb` (servers), `mscc-init-gui_..._all.deb`, `mscc-ui_..._amd64.deb`, and the optional `mscc-firmware_..._all.deb`.

Use the right kit. Never install `_arm64` packages on an Ubuntu PC. The only `_all` files are shared. Don't install `pulseaudio` on Ubuntu: Ubuntu already uses PipeWire, and MSCC works with it.

You install MSCC by pasting a few commands into a **Terminal**. In the Files app, open the install folder, right-click an empty spot and choose **Open in Terminal**. When a command starts with `sudo`, type your login password (nothing shows while you type; that's normal).

Run these commands one at a time, **in this order**, waiting for each to finish. The `*` means "whatever version is in the folder". On a computer where a package is already installed, `apt` reports "... is already the newest version" (as in the pictures); that is fine.

1. **Permission for the radio's serial and audio ports**, then **log out and log back in** (or restart) and open the Terminal in the install folder again:

```
sudo usermod -aG dialout,audio,plugdev "$USER"
```

```
stew@stew-HP-Notebook:~$ sudo usermod -aG dialout,audio,plugdev "$USER"  
[sudo: authenticate] Password:
```

usermod (asks for your password)

2. **Audio library (PortAudio):**

```
sudo apt install -y ./mscc-portaudio*_amd64.deb
```



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```
stew@stew-HP-Notebook:~/Documents/GitHub/mscc-station/installers/linux$ sudo apt install -y ./mscc-portaudio*_amd64.deb
Note, selecting 'mscc-portaudio' instead of './mscc-portaudio_19.8.2_amd64.deb'
mscc-portaudio is already the newest version (19.8.2).
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 59
```

mscc-portaudio installed

3. **Servers.** Run `install-mscc.sh` as yourself, not with `sudo`; it asks for your password when it needs it. It installs the servers and the virtual serial ports used for CAT.

```
chmod +x install-mscc.sh
./install-mscc.sh
```

```
stew@stew-HP-Notebook:~/Documents/GitHub/mscc-station/installers/linux$ chmod +x install-mscc.sh
stew@stew-HP-Notebook:~/Documents/GitHub/mscc-station/installers/linux$ ./install-mscc.sh
mscc installer
package: /home/stew/Documents/GitHub/mscc-station/installers/linux/mscc_1.0.50_amd64.deb
staging: /tmp/mscc-apt-install-17427.deb (world-readable so apt does not warn)
reinstall: 0
... (package list download lines omitted) ...
78 packages can be upgraded. Run 'apt list --upgradable' to see them.
Note, selecting 'mscc' instead of '/tmp/mscc-apt-install-17427.deb'
mscc is already the newest version (1.0.50).
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 78

Install finished. Next steps (in order):
1. Log out/in (or reboot) – enables groups + virtual digi audio service
2. pactl list short sinks | grep Virtual # if empty: mscc-virtual-audio
3. mscc-init # digi speaker=VirtualA digi mic=VirtualB.monitor
4. mscc start
Audio: install mscc-portaudio*_amd64.deb first (Pulse+ALSA at /usr/local)
stew@stew-HP-Notebook:~/Documents/GitHub/mscc-station/installers/linux$
```

install-mscc.sh: start and end of the output, with its next-steps list

4. MSCC Init:

```
sudo apt install -y ./mscc-init-gui*_all.deb
```

```
stew@stew-HP-Notebook:~/Documents/GitHub/mscc-station/installers/linux$ sudo apt install -y ./mscc-init-gui*_all.deb
Note, selecting 'mscc-init-gui' instead of './mscc-init-gui_1.0.13_all.deb'
mscc-init-gui is already the newest version (1.0.13).
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 78
```

mscc-init-gui installed

5. MSCC UI:

```
sudo apt install -y ./mscc-ui*_amd64.deb
```

```
stew@stew-HP-Notebook:~/Documents/GitHub/mscc-station/installers/linux$ sudo apt install -y ./mscc-ui*_amd64.deb
Note, selecting 'mscc-ui' instead of './mscc-ui_0.6.72_amd64.deb'
mscc-ui is already the newest version (0.6.72).
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 78
stew@stew-HP-Notebook:~/Documents/GitHub/mscc-station/installers/linux$
```

mscc-ui installed

6. Radio firmware files (optional, needed for Appendix A):



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```
sudo apt install -y ./mscc-firmware*_all.deb
```

```
stew@stew-HP-Notebook:~/Documents/GitHub/mscc-station/installers/linux$ sudo apt install -y ./mscc-firmware*_all.deb
[sudo: authenticate] Password:
Note, selecting 'mscc-firmware' instead of './mscc-firmware_1.0.1_all.deb'
mscc-firmware is already the newest version (1.0.1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 78
stew@stew-HP-Notebook:~/Documents/GitHub/mscc-station/installers/linux$
```

mscc-firmware installed

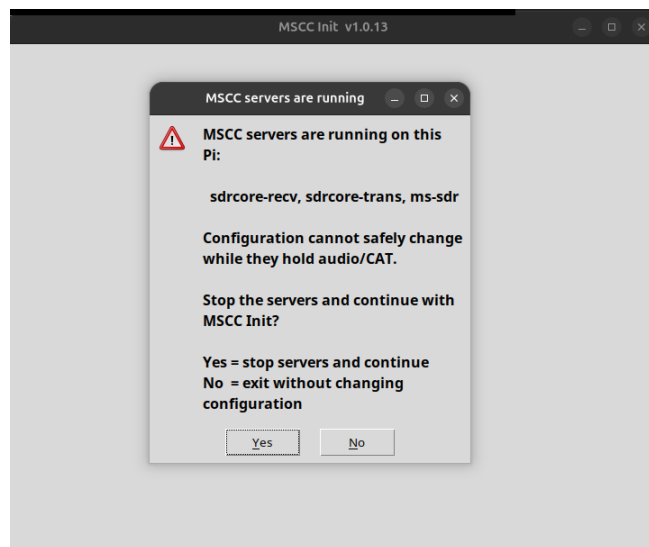
7. Run **MSCC Init** (2.2), then plug in the radio, power it on, and start **MSCC UI** (2.4).

Only one copy of MSCC UI should be open at a time. The window remembers its size and position, and your frequencies, modes, filters and display settings are remembered between sessions.

2.2 MSCC Init: setting up this computer

Run **MSCC Init** once after installing, and again if you change your speaker, microphone or radio. It has four steps, then a review page. Use **Next** → and ← **Back**; **Cancel** leaves without changing anything. The heading says “configure this Pi” even on an Ubuntu PC; that’s normal.

If the servers are running, MSCC Init asks to stop them first. **Yes** stops them and continues (“Servers stopped. Continuing with configuration.”); **No** exits without changing anything.

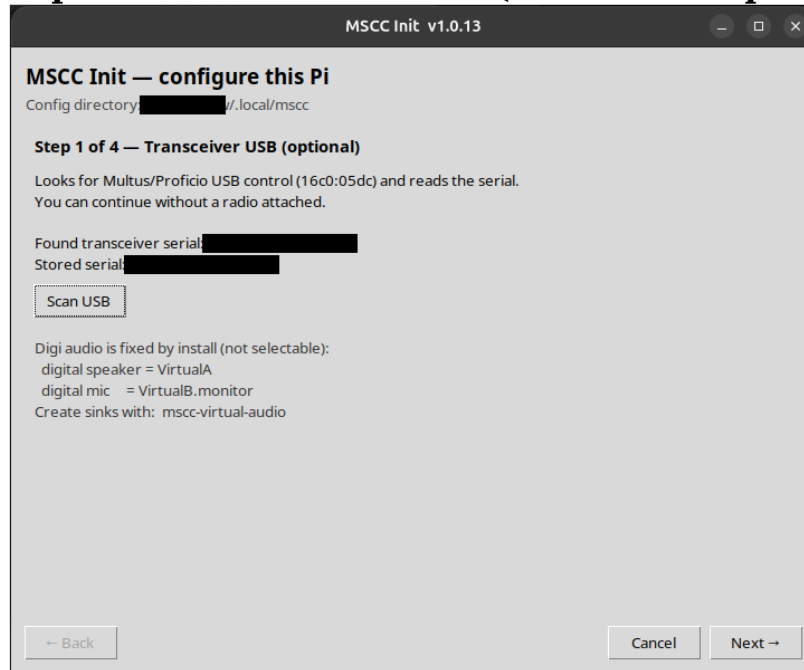


MSCC Init: servers are running. Yes stops them and continues; No exits

Step 1 of 4: Transceiver USB (optional). Click **Scan USB** with the radio plugged in and on. MSCC Init shows the radio’s serial number (“Found transceiver serial”) and the one already saved. The fixed digital audio devices (**VirtualA** and **VirtualB.monitor**) are shown too; you don’t pick them.



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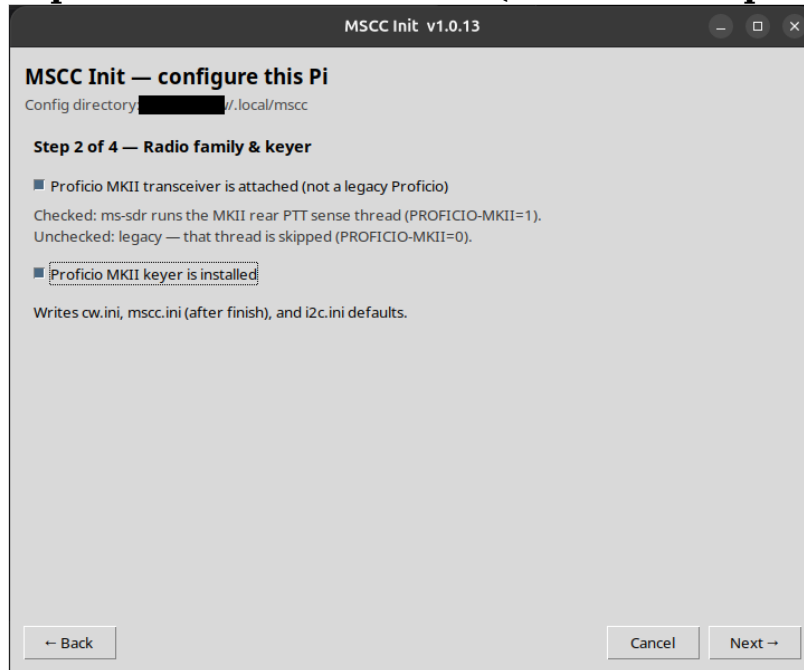
Step 1 after Scan USB (serial number blacked out)

Step 2 of 4: Radio family and keyer.

1. **Proficio MKII transceiver is attached.** Check for an MKII radio, uncheck for a Legacy radio.
2. **Proficio MKII keyer is installed.** Check to use the radio's built-in keyer (paddle in the Key jack). Uncheck if you use your own external keyer.



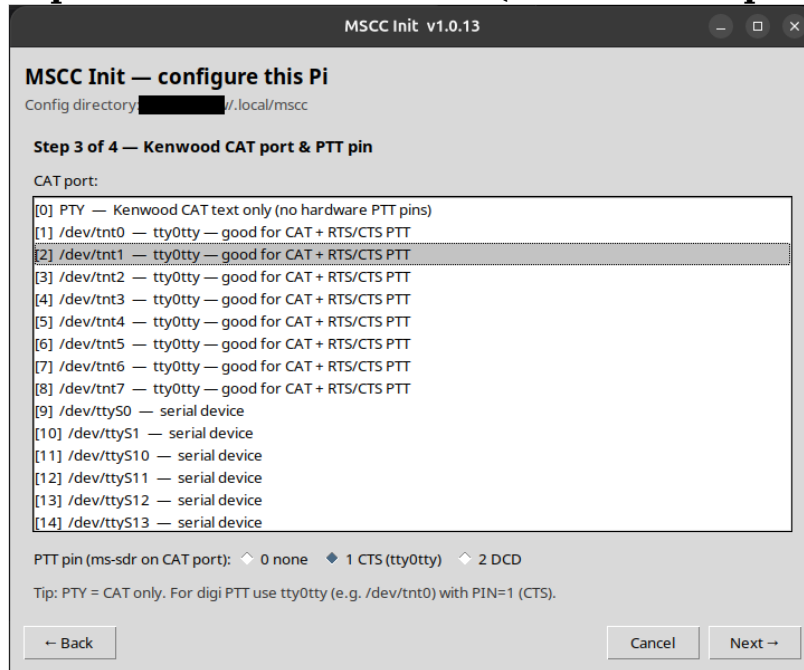
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Step 2: radio family and keyer

Step 3 of 4: Kenwood CAT port and PTT pin. The serial port the servers use for CAT (logging and digital programs). Pick one of the /dev/tnt ports; they come in pairs (tnt0 with tnt1). The servers use the end you pick and your digital program uses the other end. **PTT pin: 1 CTS (tty0tty)**. SETTINGS in MSCC UI (2.12) shows which end is which.

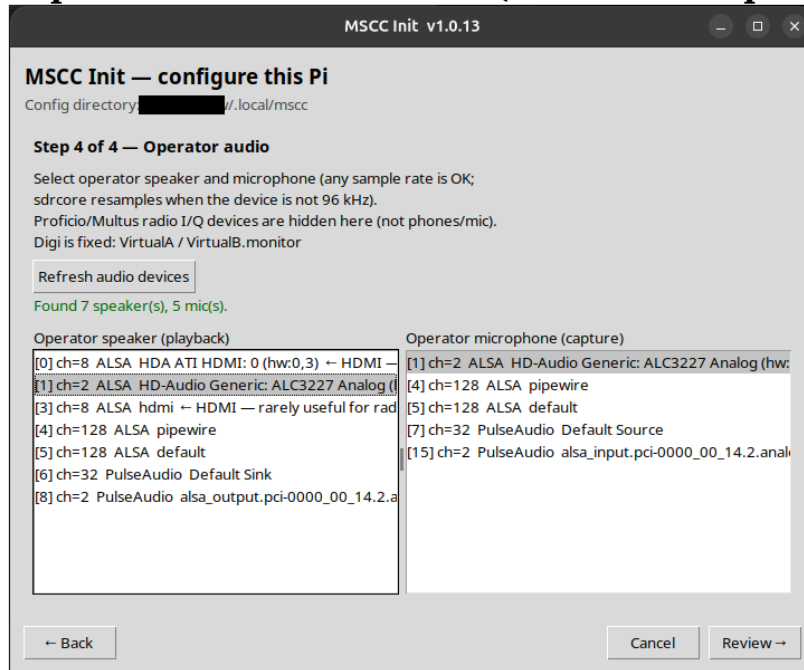
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Step 3: CAT port /dev/tnt1 and PTT pin 1 CTS

Step 4 of 4: Operator audio. Pick the **Operator speaker** and **Operator microphone** for voice and for hearing the radio: the computer's built-in audio or your headset, not an HDMI output. **Refresh audio devices** rescans after you plug in a headset. Click **Review →**.

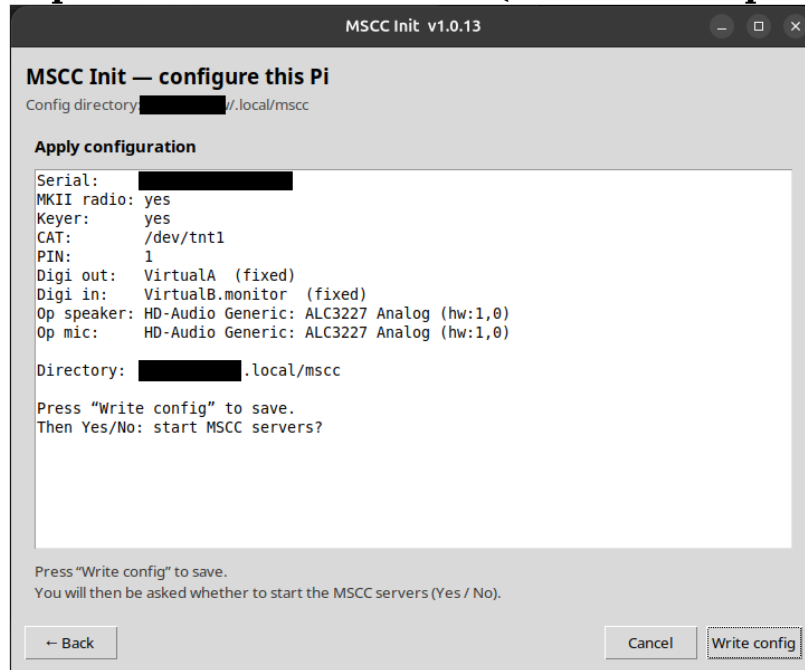
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Step 4: operator speaker and microphone

Review. Check your choices and click **Write config**. MSCC Init then asks “**Start MSCC servers now?**” **Yes** starts them (“MSCC servers started.”); **No** finishes without starting them (MSCC UI’s Launch can start them, 2.4).

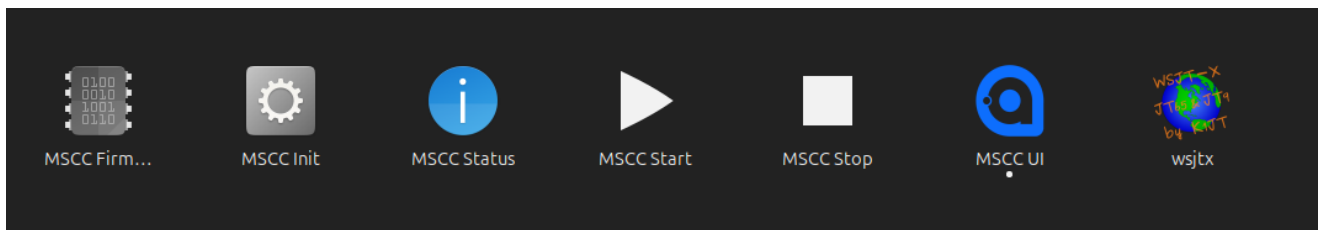
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Review page: Write config

2.3 The MSCC menu, and MSCC Start, Stop and Status

After installing, the MSCC entries appear with your other programs (Ubuntu: **Show Apps** at the bottom of the dock). Right-click one and choose **Pin to Dash** to keep it on the dock.



The MSCC entries in the Ubuntu app grid

1. **MSCC Firmware Upload.** Loads new firmware into the radio (Appendix A).
2. **MSCC Init.** Sets up this computer (2.2).
3. **MSCC Status.** Opens a Terminal window with a health check.
4. **MSCC Start.** Starts the servers (“Servers started (recv, trans, ms-sdr).”).
5. **MSCC Stop.** Stops the servers (“Servers stopped.”). Stop them before you run MSCC Init or update the firmware.
6. **MSCC UI.** The operating window (2.4 onward).
7. **wsjtx.** WSJT-X, if installed (Appendix F).

The same commands work in a Terminal: `mscc start`, `mscc stop`, `mscc restart`, `mscc status`.



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MSCC Status reports whether each server is running, the settings MSCC Init wrote, whether your audio devices and CAT port were found, and the radio's USB control device. Lines start with **OK**, **WARN** or **FAIL**; the **Summary** should read "**MSCC status: OK**" and "**FAIL=0 WARN=0**". A picture of this window is very useful with a problem report. Press Enter to close it.

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```
MSCC status - /home/stew/mscc
sdrcore-recv: stopped
sdrcore-trans: stopped
ms-sdr: stopped
MSCC status report - /home/stew

=== Servers ===
sdrcore-recv: stopped
sdrcore-trans: stopped
ms-sdr: stopped

=== Binaries ===
MSCC_DIR=/home/stew/mscc
OK sdrcore-recv
OK sdrcore-trans
OK ms-sdr
OK mscc.sh

=== PortAudio (library used by cores) ===
libportaudio.so.2 => /usr/local/lib/libportaudio.so.2 (0x00007b427ccb1000)
OK linked against /usr/local (mscc-portaudio)

=== Config ($HOME/.local/mscc) ===
config dir: /home/stew/.local/mscc
OK operator speaker: HD-Audio Generic: ALC3227 Analog
OK operator mic: HD-Audio Generic: ALC3227 Analog
OK digital speaker: VirtualA
OK digital mic: VirtualB.monitor
OK CAT / comm: COMM_PORT_NAME=/dev/tty1 PIN=1
file: /home/stew/.local/mscc/comm-port.ini
OK mscc.ini: se [REDACTED] ports 8889/8888

=== PortAudio devices vs config ===
PortAudio sees 22 devices; host APIs: ALSA, PulseAudio
OK Proficio/Multus present: pa=2 api=ALSA in=2 out=2 name='Proficio: USB Audio (hw:2,0)'
```

MSCC Status, top of the report (radio serial number and host name blacked out)

```
OK Operator speaker: matched pa=1 api=ALSA in=2 out=2 rate=44100 name='HD-Audio Generic: ALC3227 Analog (hw:1,0)'
```

MSCC Status, end of the report: Summary OK

2.4 Connecting to the radio

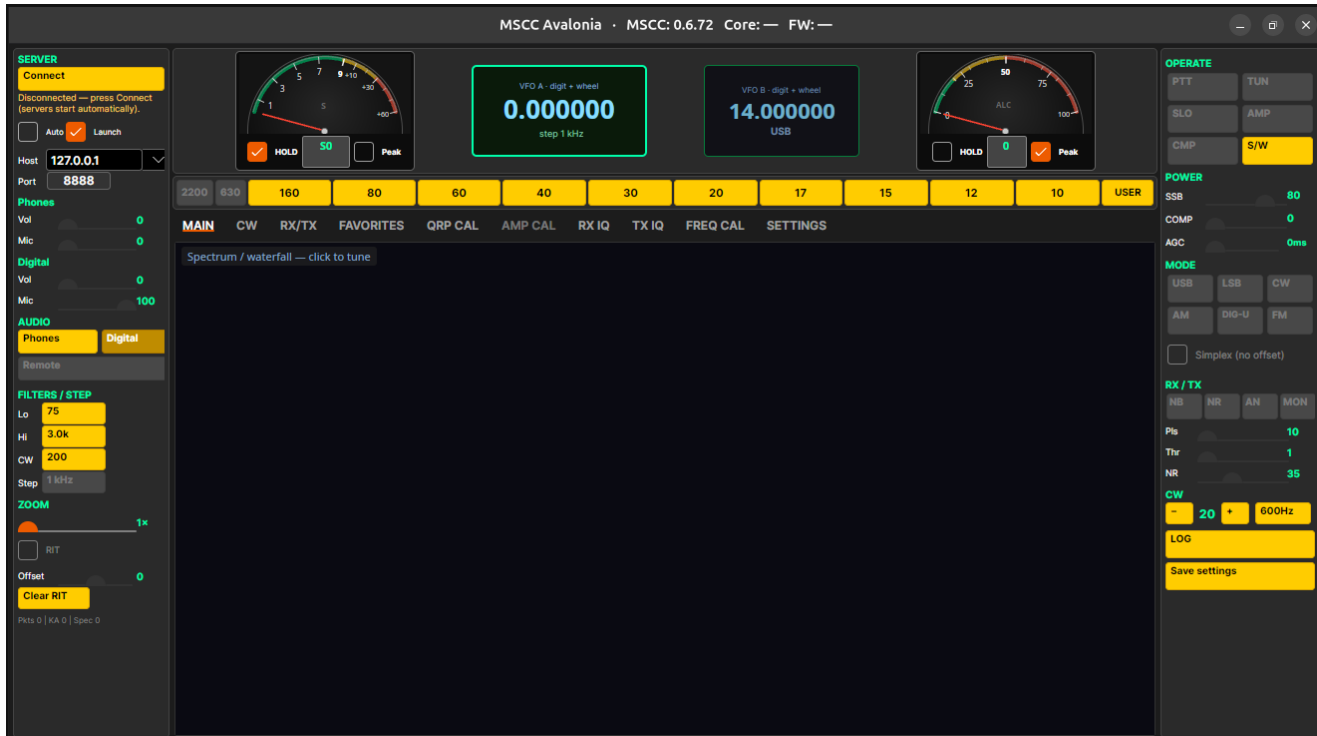
On the left rail, under **SERVER**:

- **Launch** checked (normal for local operation): **Connect** also starts the servers on this computer if they aren't running, and **Disconnect** stops them again. Launch works only with Host **127.0.0.1**, and only while disconnected.
- **Launch** unchecked: **Connect** only connects, to servers you started yourself (MSCC Start).
- **Auto** presses Connect for you when MSCC UI opens.



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- **Host** is **127.0.0.1** for local operation. A change takes effect on the next Connect. ▼ lists hosts you used before.
- **Port** is **8888** unless you changed it on the servers.



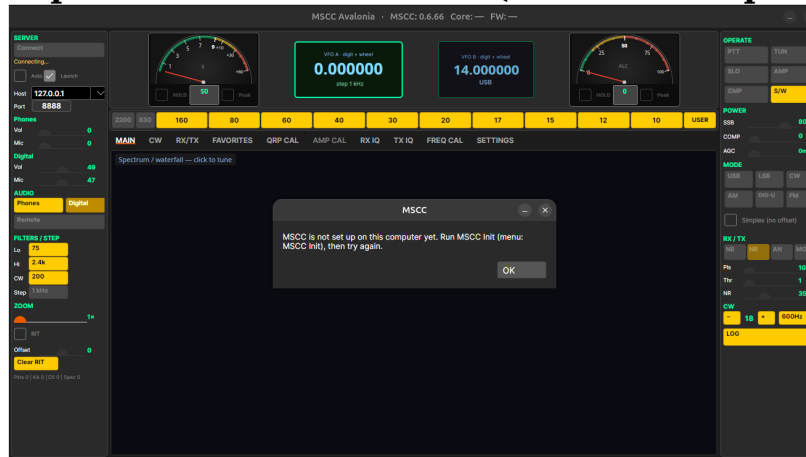
MSCC UI just opened, not connected

Before you connect, the title bar shows “Core: — FW: —”, the VFOs read 0.000000 and the spectrum is empty. The line under the button tells you what Connect will do: “Disconnected — MSCC Start, then Connect.” (Launch off) or “Disconnected — press Connect (servers start automatically).” (Launch on).

While the servers start, the status line reads “**Starting servers...**” then “**Waiting for servers...**”, and Ubuntu shows a desktop notification, “**Servers started (recv, trans, ms-sdr).**” When connected, the button reads **Disconnect**, the status line reads “**Connected — spectrum OK**”, and the title bar shows the versions and your radio model, for example (yours will differ): MSCC Avalonia · MSCC: 0.6.72 Core: 3.176 FW: 3.232 Proficio MKII PTT

If MSCC Init hasn’t been run, Connect shows “**MSCC is not set up on this computer yet.**” It’s a warning: click **OK** and MSCC UI carries on, but run MSCC Init (2.2) when you can.

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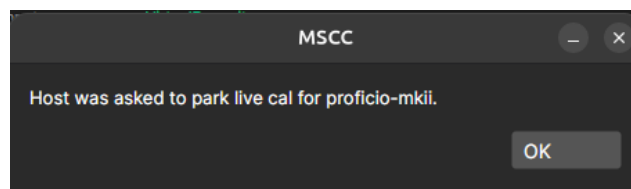
Warning when MSCC Init has not been run

Disconnect turns transmit off, ends any calibration that is running, and ends the session. If MSCC UI started the servers (Launch), it stops them too (“Servers stopped.”). Closing the window does the same. Servers you started with MSCC Start keep running.

If the status line shows **“WARNING: keep-alive lost”**, MSCC UI has stopped hearing from the servers. Check the radio’s power and USB, then Disconnect and Connect again (Appendix E).

Save settings. On the right rail under **LOG**, press **Save settings** to park this radio’s live TX IQ, QRP and AMP calibration on this computer (the one the radio is plugged into), filed by radio line under `~/ .local/mscc/cal/` (Appendix D).

- When connected, MSCC confirms: “Host was asked to park live cal for proficio-mkii.” (your radio line is shown).

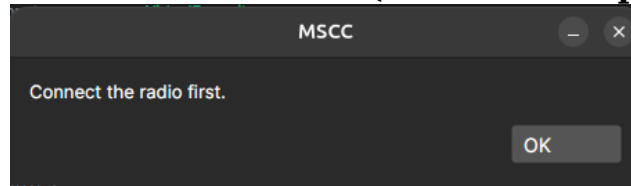


Save settings: parked on the host

- If the radio is not connected yet, MSCC shows “Connect the radio first.”

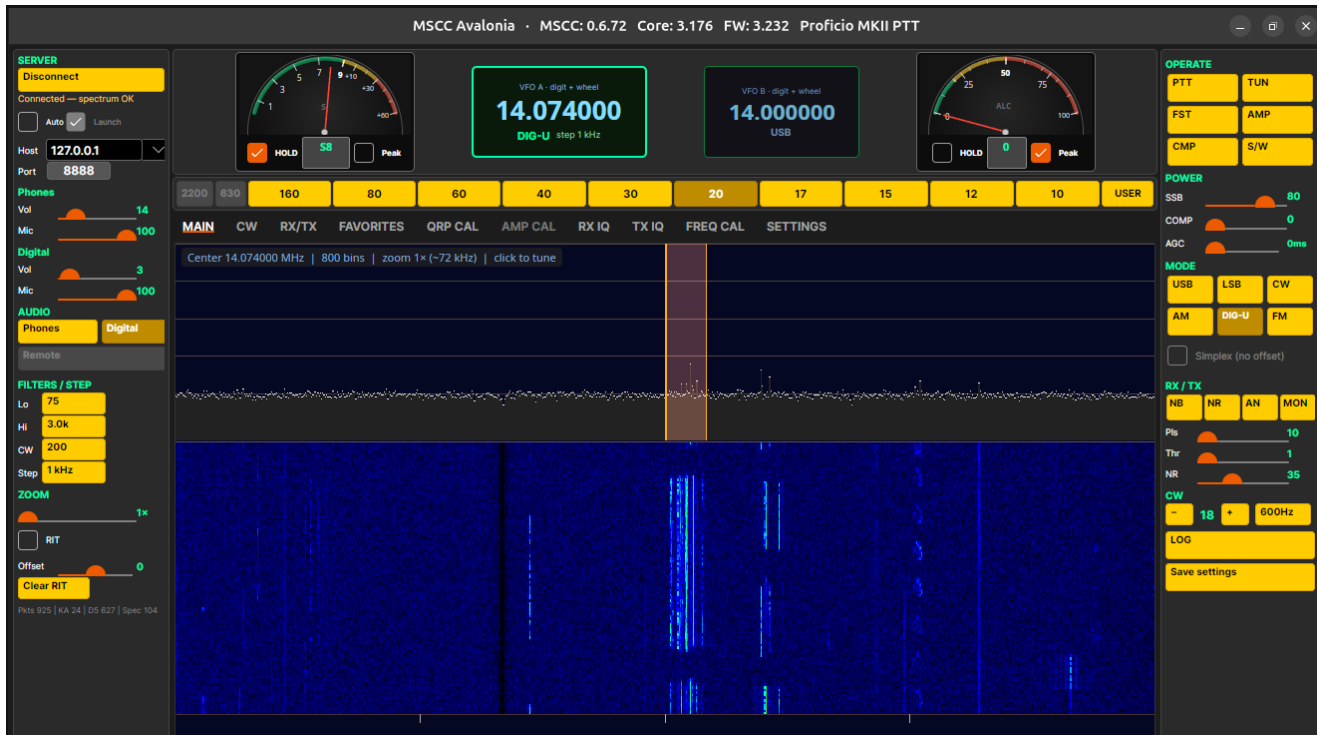


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Save settings: connect the radio first

2.5 The Main Window



The main window (enlarged views follow with each list)

VFO A on 20 m in DIG-U. The list below names each control, left rail first, then the center column, then the right rail, top to bottom.

The main window has three columns: the **left rail** (connection, audio, filters, zoom and RIT), the **center** (meters and VFOs, band buttons, then the tabs) and the **right rail** (operating controls). Both rails stay visible on every tab.

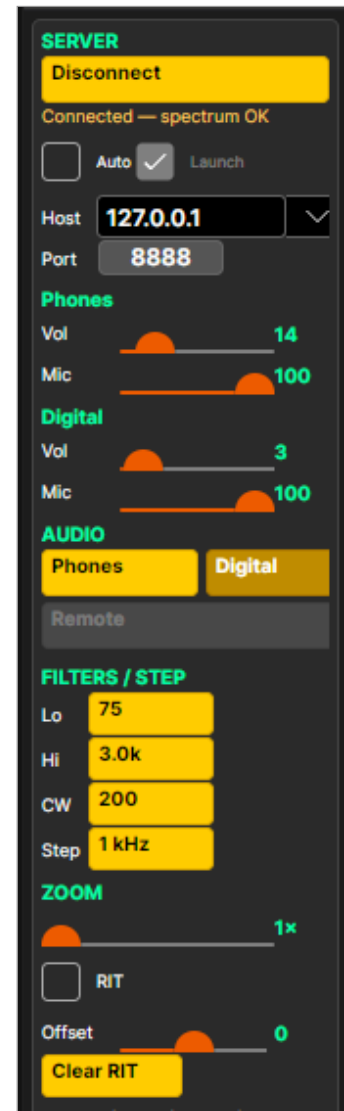
Button colors: bright gold = off; darker gold (brown) = on or selected; **red** = transmitting or a transmit switch that is on (PTT, TUN, calibration TX); gray = not available right now.

Mouse wheel: over any slider, one notch moves the slider one small step.

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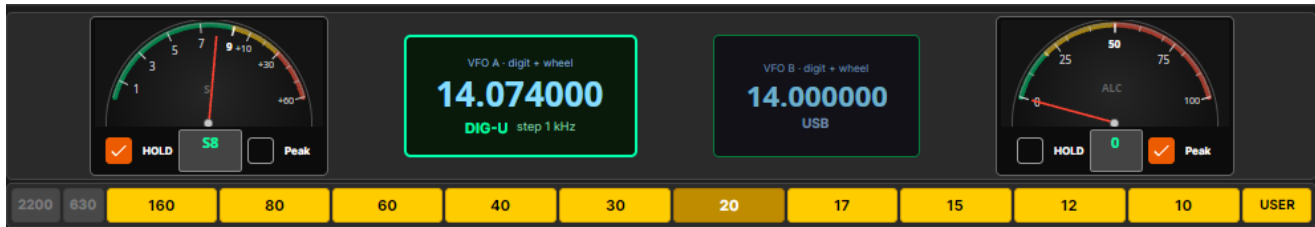
Left rail

1. **Connect / Disconnect** (SERVER), with a status line under it. See 2.4.
2. **Auto** checkbox. See 2.4.
3. **Launch** checkbox. See 2.4.
4. **Host** box and recent-hosts ▼. **127.0.0.1** for local operation.
5. **Port**. Normally 8888.
6. **Phones Vol** (0–100). Headphone and speaker volume.
7. **Phones Mic** (0–100). Microphone gain.
8. **Digital Vol** (0–100). Receive audio level sent to your digital program.
9. **Digital Mic** (0–100). Transmit audio level from your digital program.
10. **Phones** (AUDIO). Use your speaker or headphones and microphone.
11. **Digital** (AUDIO). Use the digital audio path (WSJT-X and similar). Choosing **DIG-U** switches to Digital for you; leaving DIG-U puts back what you had.
12. **Remote**. Grayed out for local operation (Host 127.0.0.1). See the Remote guide.
13. **Lo** (low cut). 500 → 300 → 200 → 100 → 75 Hz.
14. **Hi** (high cut). DIG-U: 5.5k → 4.0k → 3.0k → 2.7k → 2.4k → 1.4k → 1.0k. Other modes: 5.5k → 4.0k → 3.0k → 2.7k → 2.4k.
15. **CW** (CW bandwidth). 1.8 kHz → 400 Hz → 200 Hz.
16. **Step**. Mouse-wheel tuning step: 10 Hz → 100 Hz → 1 kHz → 10 kHz → 100 kHz. Double-clicking VFO A also changes it.
17. **ZOOM** (1×–4×). Magnifies the spectrum and waterfall around your frequency (also in S/W).
18. **RIT** checkbox. Turns receive offset on.
19. **Offset** slider (–500 to +500 Hz).
20. **Clear RIT**. RIT off and offset back to 0.
21. **Counters** (“Pkts ... | KA ... | DS ... | Spec ...”). Data arriving from the servers; useful in problem reports.



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Meters and VFOs



Meters and VFOs

22. **S-meter**, with **HOLD**, the reading box, and **Peak**. **HOLD** (on by default) makes the needle fall back slowly so you can read it. **Peak** holds the highest reading. The reading box shows the S reading.
23. **VFO A**. Click to make it active (bright border). Wheel over a digit tunes that digit (digits to its right go to zero); wheel over the box but not a digit tunes by **Step**; double-click to change Step. The mode shows under the frequency.
24. **VFO B**. Works the same way; each VFO remembers its own band. In FM repeater mode VFO B shows the transmit frequency.
25. **ALC meter**, with **HOLD**, the reading box, and **Peak**. Transmit ALC level; rests at zero on receive. HOLD slows the needle's fall; Peak holds the highest reading.

Band buttons and tabs

26. **Band buttons**: 2200, 630, 160, 80, 60, 40, 30, 20, 17, 15, 12, 10 and **USER**. Bands your radio can't use are grayed out (Proficio and Ultimur: 160–10 m; Geminus: 2200 and 630 m). Each band returns to the frequency, mode and filters you last used there. **USER** steps through WWV 5, 10, 15 and 20 MHz, CHU 3.330 and 7.850 MHz, RWM 9.996 MHz, and your own last general-coverage frequency; these are good references for FREQ CAL.
27. **Tab strip**: MAIN, CW, RX/TX, FAVORITES, QRP CAL, AMP CAL, RX IQ, TX IQ, FREQ CAL, SETTINGS. The calibration tab that doesn't match the AMP setting is gray (QRP CAL and TX IQ need AMP off; AMP CAL needs AMP on).
28. **Spectrum and waterfall** (MAIN tab). Left-click a signal to tune to it. The top line shows the center frequency, resolution ("800 bins"), zoom and "click to tune". The shaded band is your receive passband. Colors, levels and **SPECTRUM RESOLUTION** are in the S/W window (2.13).

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Right rail

29. **PTT.** Transmit on/off; red while on. Voice modes only; in CW the paddle or keyer keys the radio. PTT and TUN lock while a digital program or hardware PTT has control (“Server owns TX — PTT/TUN locked”).
30. **TUN.** Steady carrier at **Tune Power** for your tuner or amplifier.
31. **SLO / MED / FST** (AGC). Each click changes the AGC speed.
32. **AMP.** Switches between the two power tables. **AMP off** = QRP up to the radio’s full 5 W (QRP CAL table). **AMP on** = the AMP CAL table (2.8), to drive your amplifier correctly. Red while on.
33. **CMP.** Speech compressor on/off.
34. **S/W.** Opens the Spectrum / Waterfall Controls window (2.13).
35. **POWER slider** (0–100 %). The label follows what you’re doing: TUNE, CW, AM, FM or SSB (LSB, USB, DIG-U). Same settings as on RX/TX. Not watts: a percentage of your QRP CAL level.
36. **COMP** (0–24). Compressor amount. Phones path only.
37. **AGC** (0–1000 ms). AGC fast release.
38. **Mode buttons:** **USB, LSB, CW, AM, DIG-U, FM.** DIG-U is upper sideband for digital modes (Digital path, extra 1.4k/1.0k Hi filters). FM transmits 100 kHz below receive by default (repeater split).
39. **Simplex (no offset).** FM only: transmit on the receive frequency.
40. **NB / NR / AN / MON.** Noise blanker, noise reduction, automatic notch, and monitor, each on/off.
41. **Pls** (10–510). Noise blanker pulse width.
42. **Thr** (1–100). Noise blanker threshold.
43. **NR** (0–100). Noise reduction strength.
44. **CW – / speed / +.** Keyer speed, 5–60 WPM.
45. **CW pitch.** 400 → 600 → 800 → 1000 Hz.
46. **LOG.** Opens the Debug Log window (2.14).
47. **Save settings.** Parks this radio’s TX IQ, QRP and AMP calibration on this computer (2.4).



▲ WARNING: Don’t overdrive your amplifier. Multus amplifiers give full output with about 75 % of the radio’s power. Feeding the full 5 W into the amplifier overdrives it. Always turn **AMP on** when the amplifier is in line, and calibrate AMP CAL first (2.8).



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2.6 CW Tab

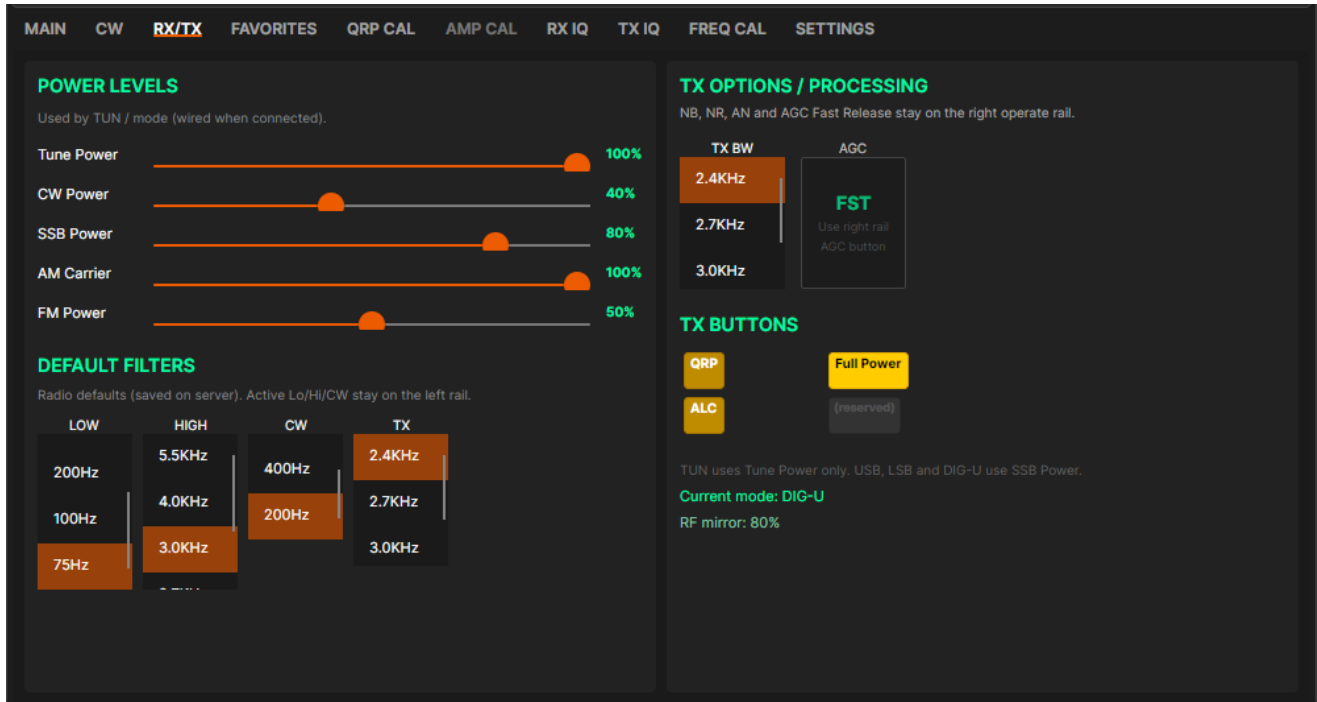
CW tab

1. **Use external electronic keyer (legacy).** Check if you key the radio with your own external keyer, or have a Legacy radio. The built-in keyer controls gray out (HOLD still works). The servers read this only when they start: Disconnect and Connect again (with Launch), or MSCC Stop and MSCC Start.
2. **KEYER MODE.** STRAIGHT, IAMBIC-A (default) or IAMBIC-B.
3. **SPACING.** ELEMENT or LETTER.
4. **PADDLE.** NORMAL or REVERSE (swaps dot and dash).
5. **WEIGHT.** 25, 50 (default) or 75.
6. **PITCH.** 400, 600, 800 or 1000 Hz. Same as CW pitch on the right rail.
7. **SPEED** – / +. 5–60 WPM. Same as the right rail.
8. **FARNSWORTH** – / +. Off, or 5–60 WPM. Extra space between letters and words **when playing CQ memories only**.
9. **HOLD** – / +. 1–500 ms in steps of 10 (default 100). How long the radio stays in transmit after you stop keying.
10. **POTENTIAL / QSK.** Amplifier PIN-diode T/R switching. Checked, the amplifier can switch with each dit/dah. Unchecked, it holds TX until HOLD runs out. Full QSK is not supported.
11. **CQ MEMORY slots 0–3.** Each slot has a text box (up to 48 characters), a character count, **R** (store in the radio's keyer) and **P** (play once; touch the paddle to stop). R and P work only while connected.

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See also Appendix B.

2.7 RX/TX Tab



RX/TX tab

POWER LEVELS

1. **Tune Power** (0–100 %). Drive for TUNE only.
2. **CW Power** (0–100 %).
3. **SSB Power** (0–100 %). LSB, USB and DIG-U.
4. **AM Carrier** (0–100 %).
5. **FM Power** (0–100 %).

DEFAULT FILTERS

6. **LOW, HIGH, CW, TX**. The radio's stored default filters (saved on the servers). Day-to-day filters are the Lo, Hi and CW buttons on the left rail.

TX OPTIONS / PROCESSING

7. **TX BW**. Transmit bandwidth: 2.4k, 2.7k, 3.0k or 5.5k (scroll the list for 5.5k). Only on this tab in MSCC UI.
8. **AGC**. Shows the current AGC speed; change it on the right rail.



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TX BUTTONS

9. **QRP.** Barefoot power, same as AMP off.
10. **Full Power.** The amplifier setting, same as AMP on (AMP CAL table).
11. **ALC.** Transmit ALC on/off. Keep ALC on, especially for digital modes.
12. **Notes.** “TUN uses Tune Power only. USB, LSB and DIG-U use SSB Power.”, the current mode and the power in use (“RF mirror”).

2.8 FAVORITES Tab

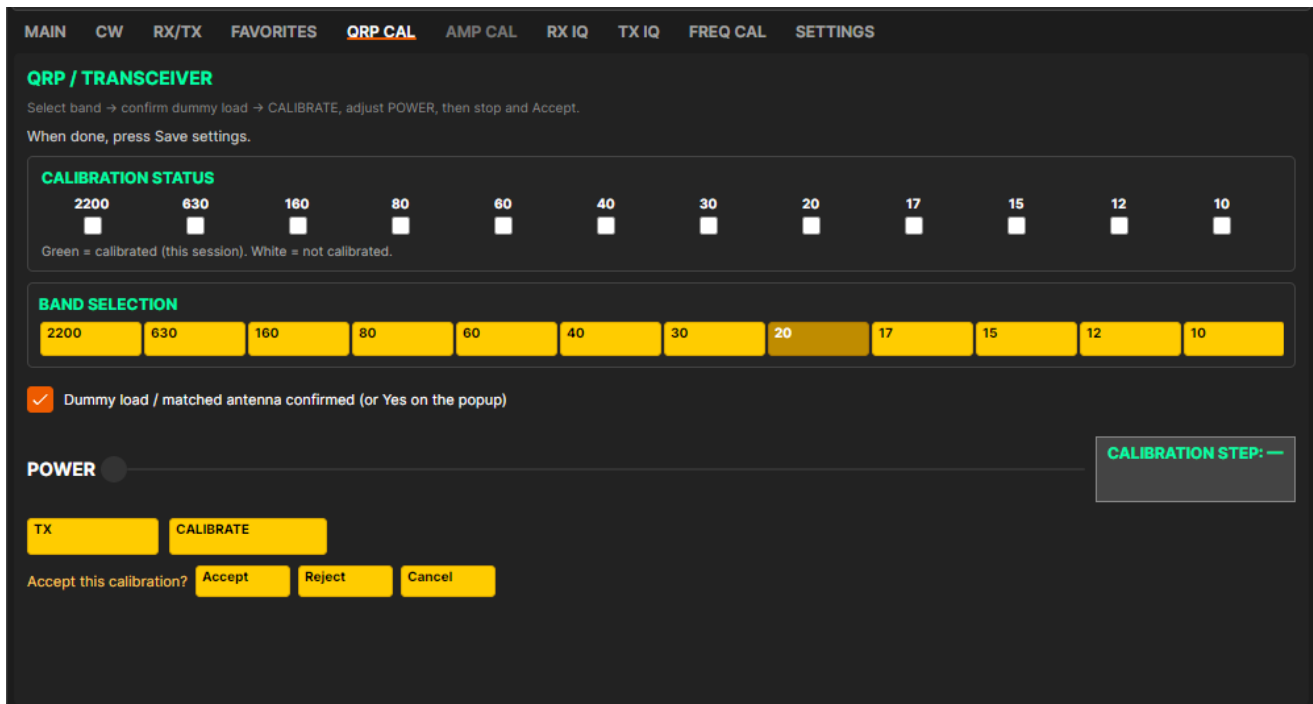
FAVORITES tab with one saved favorite

1. **Band.** Each band has its own list of favorites.
2. **Name.** Name for a new favorite (for example “20M FT8”).
3. **SAVE.** Saves the current frequency, mode, filters and VFO under this name, on the radio’s current band.
4. **RECALL.** Returns to the selected favorite.
5. **DELETE.** Removes the selected favorite.
6. **Favorites list.** Name, VFO, band, frequency, mode and filters. Click a row to select it.

Favorites are kept after you close MSCC UI (~/.config/MSCC/mscc-favorites.ini). The note at the bottom still says “Session-only (not saved to disk yet)”; that note is out of date.

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2.9 QRP CAL Tab (power calibration; AMP must be off)



QRP CAL tab, 30 m selected, waiting for Accept

This tab sets the radio's output power on each band. Your calibration is kept by the servers on this computer. The tip on the tab reminds you: **When done, press Save settings.**

Do QRP CAL first, then AMP CAL. The AMP table is built on top of the QRP level.

- You need a 50 Ω dummy load rated for 5 W or more, and a wattmeter.
- **Do not set the power above 5 W. More than 5 W causes spurious emissions.**

1. **CALIBRATION STATUS lamps.** Green = calibrated **this session**; white = not yet this session (your saved calibration is still in use).
2. **BAND SELECTION.** The radio tunes to that band's calibration frequency.
3. **Dummy load / matched antenna confirmed.** Check it, or answer **Yes** when asked.
4. **POWER slider** (0–100). Works only while CALIBRATE is on.
5. **CALIBRATION STEP.** The current setting reported by the radio.
6. **TX.** Sends a carrier at the calibration power. Turn it off before CALIBRATE.
7. **CALIBRATE.** Starts calibration (reads CALIBRATING); click again to stop, then **Accept**, **Reject** or **Cancel** on the row below.

Procedure:

1. Turn AMP off and connect the dummy load and wattmeter.



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2. Pick a band and click **CALIBRATE**. The first time in a session, MSCC asks “IS A WELL MATCHED ANTENNA OR DUMMY LOAD ATTACHED?” Click **Yes**.
3. Move **POWER** up until the wattmeter reads 5 W (not more).

The first slider move can take a few seconds. Nudge the slider up a little, wait until output begins, then set the power.

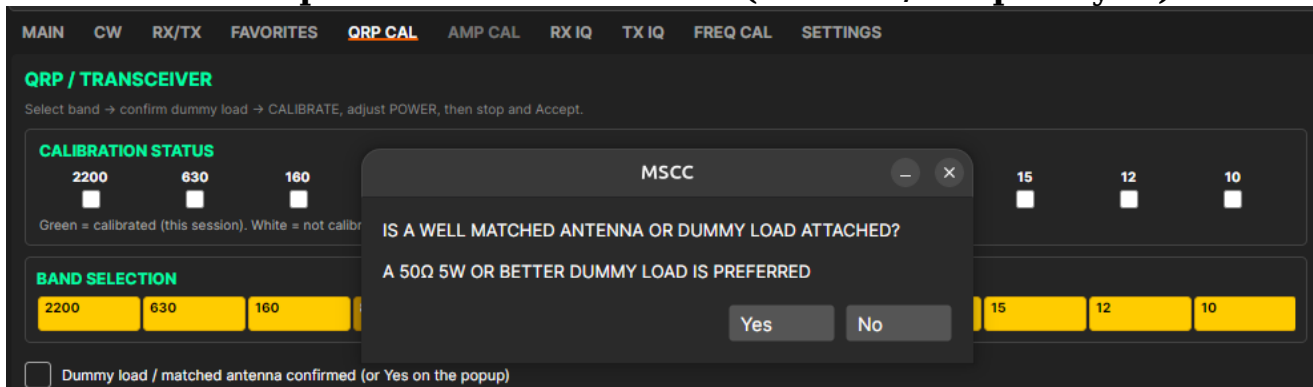
4. Click **CALIBRATE** again. **Accept** saves it and the lamp turns green; **Reject** marks the band not calibrated; **Cancel** puts back the previous setting.
5. Repeat for each band, then press **Save settings** (right rail).

The screenshot shows the 'QRP / TRANSCEIVER' interface. At the top, there are tabs: MAIN, CW, RX/TX, FAVORITES, QRP CAL (selected), AMP CAL, RX IQ, TX IQ, FREQ CAL, and SETTINGS. Below the tabs, the title 'QRP / TRANSCEIVER' is displayed in green. A subtitle reads: 'Select band → confirm dummy load → CALIBRATE, adjust POWER, then stop and Accept. When done, press Save settings.' The 'CALIBRATION STATUS' section shows a row of 12 bands: 2200, 630, 160, 80, 60, 40, 30, 20, 17, 15, 12, and 10. Each band has a small square indicator below it. A legend below the indicators states: 'Green = calibrated (this session). White = not calibrated.' The 20m band (20) has a green indicator, while all others are white. The 'BAND SELECTION' section shows the same 12 bands as yellow buttons. The 20m button is highlighted in orange. Below this, there is a checkbox labeled 'Dummy load / matched antenna confirmed (or Yes on the popup)' which is checked. A 'POWER' slider is visible, and a 'CALIBRATION STEP: —' button is on the right. At the bottom, there are 'TX' and 'CALIBRATE' buttons.

After Accept: the 20 m lamp is green

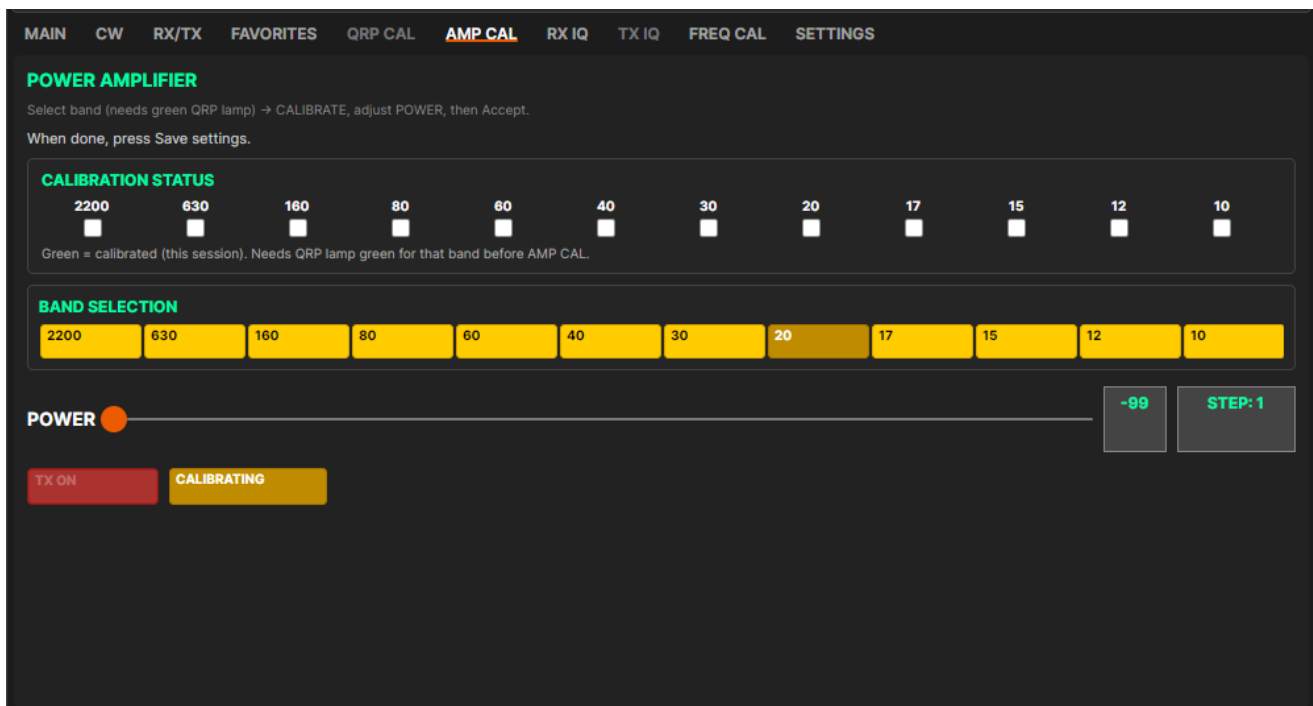
Messages: “Select a Band” (pick a band first); “Connect first” (press Connect); “TX ON. SET TX OFF BEFORE CALIBRATE”; “Finish or cancel CALIBRATE first.”; “Turn TX OFF before changing band.”

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Dummy load question, asked once per session

2.10 AMP CAL Tab (AMP must be on)



AMP CAL tab while CALIBRATING

This tab sets the **AMP power table**, the drive used when **AMP** is on, so you can switch between QRP and the amplifier without overdriving it. **When done, press Save settings.**

1. **CALIBRATION STATUS lamps.** A band's **QRP CAL** lamp must be green (this session) before MSCC lets you AMP CAL it.
2. **BAND SELECTION.**
3. **POWER slider** (–99 to 0). Works only while CALIBRATE is on; starts at –99 (lowest drive).
4. **Value** and **STEP** readouts.



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5. **TX.** Sends a tune carrier through the amplifier path.
6. **CALIBRATE.** Starts and stops calibration; then “Accept this AMP calibration?” with Accept, Reject or Cancel.

Accept this AMP calibration?

Procedure: do QRP CAL for the band first; connect the amplifier to a dummy load with a wattmeter on its output; turn **AMP** on; pick the band and click **CALIBRATE** (answer Yes to the dummy load question); raise **POWER** slowly until the amplifier just reaches rated output (the first slider move can take a few seconds); click **CALIBRATE** again and **Accept**. Then press **Save settings**.

Messages: “Select a Band”; “QRP CAL not done for this band (green lamp required).”; “Connect first”.

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2.11 RX IQ Tab (receive image balance)



RX IQ tab, ACTIVE on 40 m

First choose an amateur band on the MAIN tab (not USER).

1. **BAND** and **FREQ**. The band and frequency being balanced.
2. **START / ACTIVE**. Starts or ends the session.
3. **UP 24 kHz**. Shifts the receiver up 24 kHz to help find the image.
4. **APPLY**. Saves the value for this band.
5. **RESET ALL**. Copies the current value to **all** bands, after you confirm on the tab.
6. **I/Q OFFSET** (–200 to +200). Adjust for the smallest image.
7. **ZERO**. Sets the offset to 0 and saves it.
8. **LO FINE (Hz)**. Fine retune during the session.
9. **RESET FREQ**. Clears LO FINE.
10. **Status line**. Tells you what to do next.
11. **Spectrum**. Live view to watch the image.

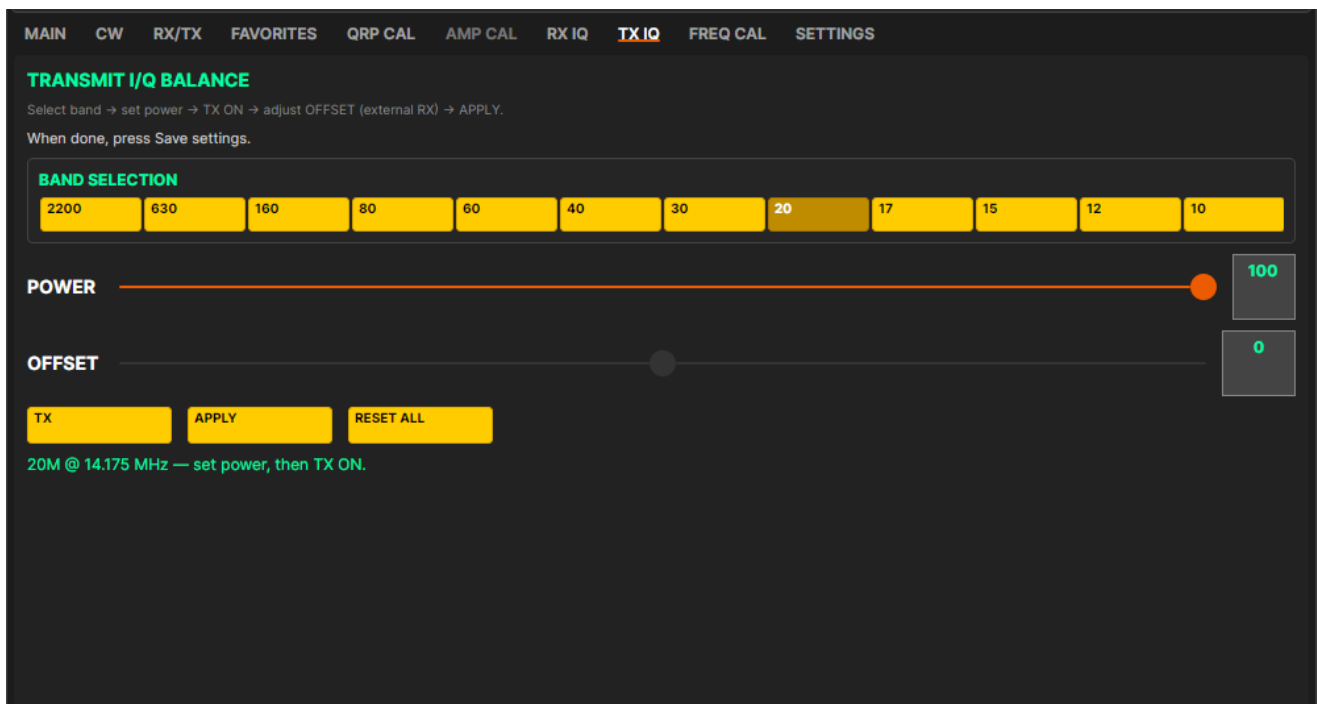
Procedure: press **START**, find the image (UP 24 kHz helps), move **I/Q OFFSET** until the image drops into the noise, then **APPLY**. The status line confirms, for example “APPLY/ZERO succeeded (IQ_OPERATION_COMPLETE).”

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After APPLY

2.12 TX IQ Tab (transmit image balance; AMP must be off)



TX IQ tab, 30 m selected

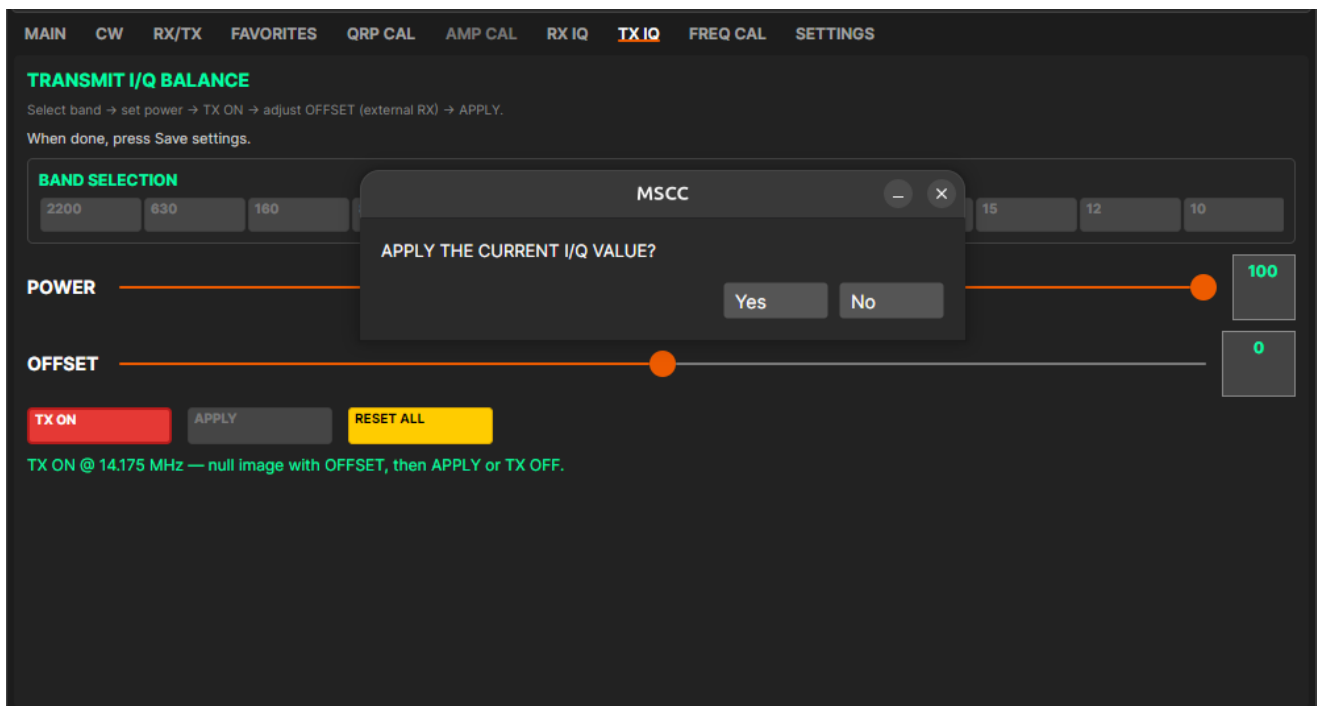


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You need a second receiver to see the transmit image. Use a dummy load and keep the second receiver's antenna well away. Tune it about 24 kHz from the displayed frequency to find the image. **When done, press Save settings.**

1. **Instructions.** Select band → set power → TX ON → adjust OFFSET (external RX) → APPLY.
2. **BAND SELECTION.** Locked while TX is on.
3. **POWER** (0–100, default 100).
4. **OFFSET** (–200 to +200). Works only while TX is on.
5. **TX / TX ON.** Starts or stops the test carrier.
6. **APPLY.** Saves the value for this band, after asking.
7. **RESET ALL.** Returns **all** bands to factory values, after you confirm. TX must be off.
8. **Status line.**

Procedure: turn AMP off, select a band, press **TX** (answer Yes to the dummy load question); move **OFFSET** until the image is weakest on the second receiver; press **APPLY** and answer **Yes** to “APPLY THE CURRENT I/Q VALUE?”; press **TX** again to stop; then **Save settings**.



TX IQ: APPLY THE CURRENT I/Q VALUE?

Pressing TX without a band gives “SELECT A BAND”; with AMP on, “TX IQ balance requires QRP mode (AMP off).”

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2.13 FREQ CAL Tab (frequency calibration)



FREQ CAL tab

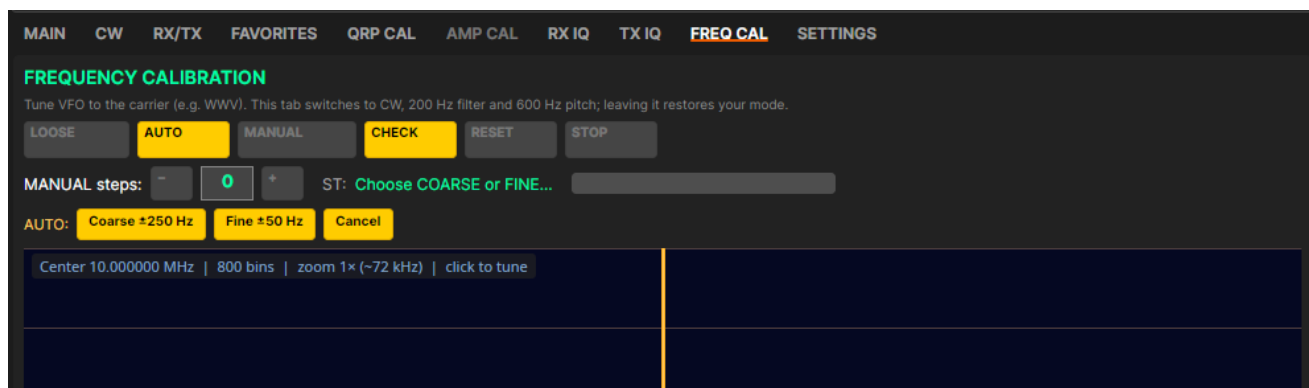
While a calibration is running, don't change tabs or settings until it finishes. MSCC won't open QRP CAL, AMP CAL or TX IQ during a run ("FREQUENCY CALIBRATION IN PROGRESS.").

Opening FREQ CAL switches to **CW**, **600 Hz** pitch and the **200 Hz** filter; leaving the tab puts back what you had. Frequency calibration is **not** part of Save settings.

1. **LOOSE / TIGHT.** How close the measurement must get. LOOSE locks more easily on a weak carrier.
2. **AUTO.** Automatic calibration against a steady carrier on your frequency. A row appears: **AUTO: Coarse ± 250 Hz / Fine ± 50 Hz / Cancel.**
3. **MANUAL.** Adjust with **MANUAL steps** - / +. Click MANUAL again; MSCC asks "**Accept manual calibration?**"
4. **CHECK.** Measures the error without changing anything (± 50 Hz range).
5. **RESET.** Returns to factory calibration, after you confirm on the tab.
6. **STOP.** Stops an AUTO or CHECK run.
7. **ST: status.** RUNNING COARSE / RUNNING FINE / CHECKING, then COMPLETED with a number in Hz, or FAILED.
8. **Progress bar.**
9. **Spectrum.** Live view.

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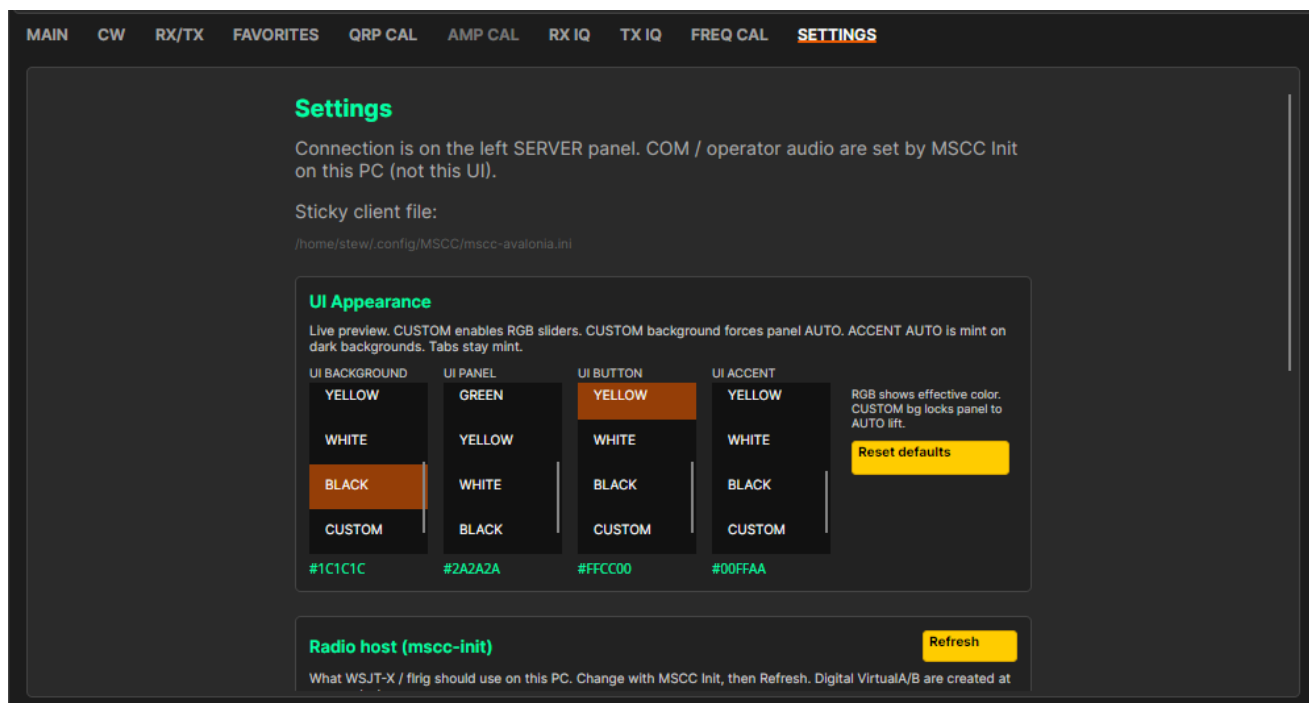
Procedure: click **USER** until you reach a strong time station (S7 or better); open **FREQ CAL**; click **AUTO** → **Coarse ±250 Hz** and wait for **AUTO COMPLETED**; then **AUTO** → **Fine ±50 Hz**; then **CHECK**. A few Hz of error is a good result.



AUTO: choose Coarse or Fine

2.14 SETTINGS Tab

The **SETTINGS** page scrolls. Unlike MSCC for Windows, the serial port and audio devices are **not** set here: MSCC Init sets them (2.2), and **SETTINGS** shows them.

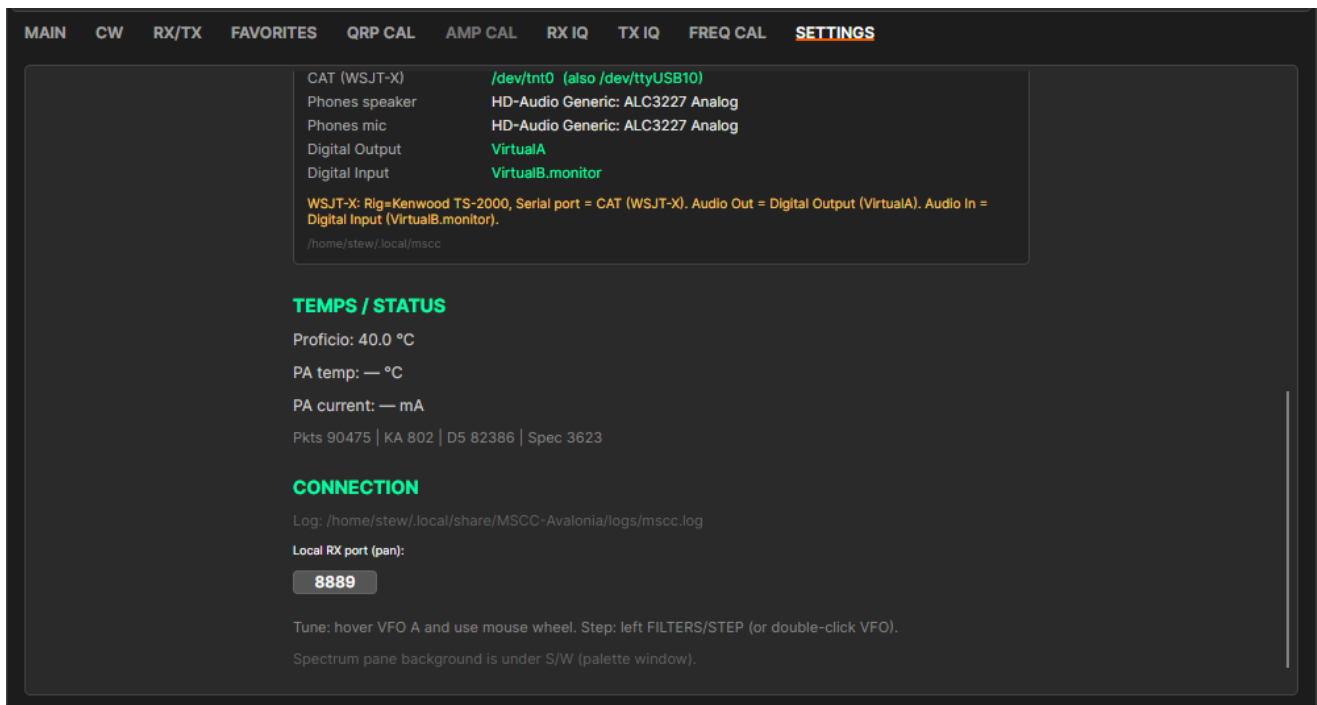


SETTINGS, top: settings file and UI Appearance

1. **Sticky client file.** Where MSCC UI keeps its settings (~/.config/MSCC/mscc-avalonia.ini).
2. **UI BACKGROUND.** YELLOW, WHITE, BLACK or CUSTOM (2.17).

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3. **UI PANEL.** Panel color.
4. **UI BUTTON.** Button color (YELLOW is the classic Multus gold).
5. **UI ACCENT.** Headings and readouts.
6. **Color codes.** The color in use, under each list.
7. **Reset defaults.**
8. **Radio host (mscc-init) and Refresh.** Read-only view of the MSCC Init settings (next picture). Change them with MSCC Init, then press Refresh.



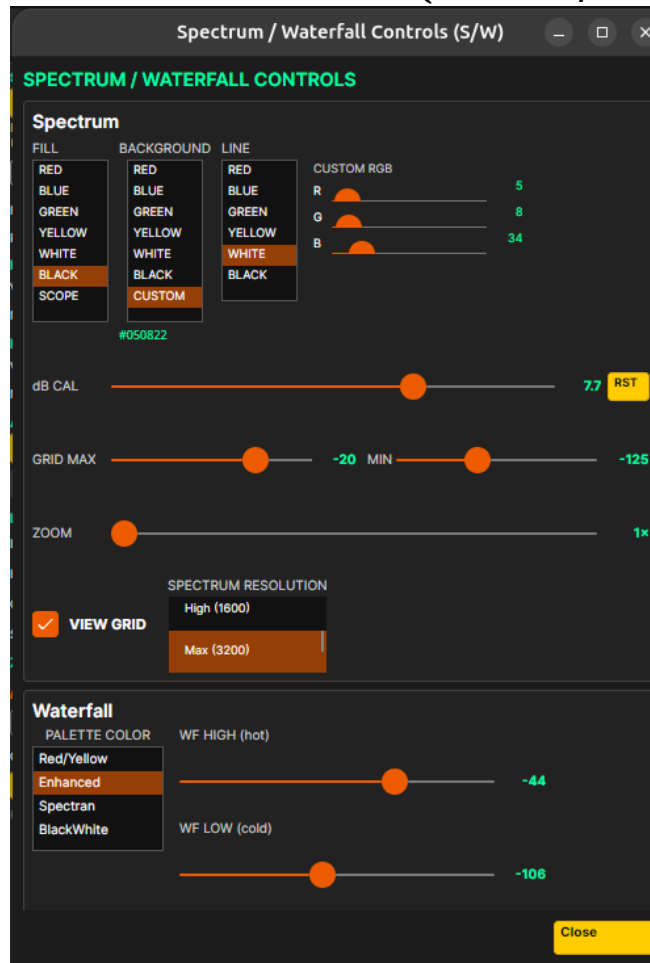
SETTINGS, scrolled: Radio host, TEMPS / STATUS and CONNECTION

9. **CAT (WSJT-X).** The port your digital program uses (the other end of the servers' pair), for example /dev/tnt0 (also /dev/ttyUSB10).
10. **Phones speaker and Phones mic.**
11. **Digital Output (VirtualA) and Digital Input (VirtualB.monitor).**
12. **WSJT-X hint** (orange). The settings to use in WSJT-X (Appendix F).
13. **Transceiver temperature** (for example "Proficio: 40.0 °C"). **PA temp** and **PA current** are reserved for later.
14. **Counters** (Pkts, KA, DS, Spec).
15. **Log** file location.
16. **Local RX port (pan).** Default 8889. Leave it alone unless told otherwise.

2.15 Spectrum / Waterfall Controls window (S/W)

Open it with **S/W** on the right rail.

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Spectrum / Waterfall Controls window (S/W)

Spectrum

1. **FILL, BACKGROUND and LINE.** Spectrum colors (FILL SCOPE = line only, the classic look). BACKGROUND **CUSTOM** shows the **CUSTOM RGB** sliders (R, G, B) and the color code under the list (2.17).
2. **dB CAL** (± 20 dB) and **RST**.
3. **GRID MAX** and **MIN**. Top and bottom of the display in dBm.
4. **ZOOM** ($1\times$ – $4\times$). Same as the left rail.
5. **VIEW GRID**. Horizontal dB grid lines.
6. **SPECTRUM RESOLUTION**. 800, High (1600) or Max (3200) bins. Higher shows finer detail. (This was PAN RESOLUTION in earlier versions.)

Waterfall

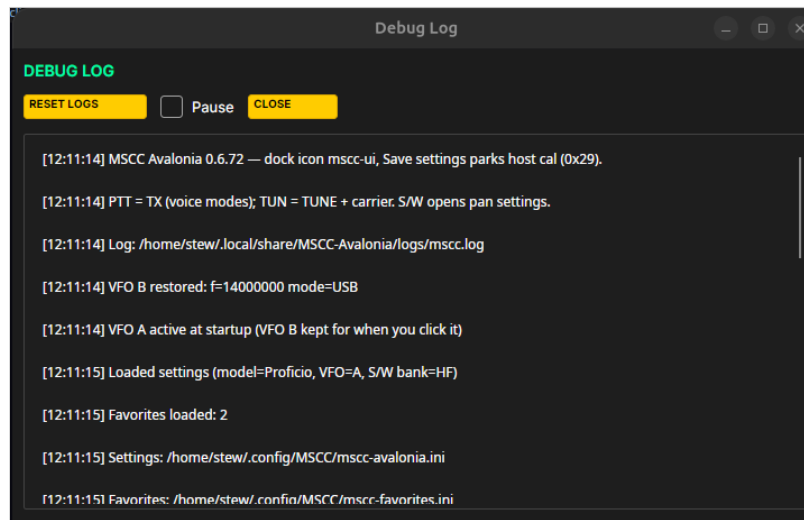
7. **PALETTE COLOR**. Red/Yellow, Enhanced, Spectran or BlackWhite.
8. **WF HIGH (hot)**. Signals this strong show in the hottest color.

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9. **WF LOW (cold).** Background level. Independent of the spectrum grid.
10. **Show waterfall.**
11. **Direction Normal.** Checked = newest line at the top scrolling down; unchecked = reversed.
12. **Close.**

2.16 Debug Log window

Open it with **LOG** on the right rail.



Debug Log window

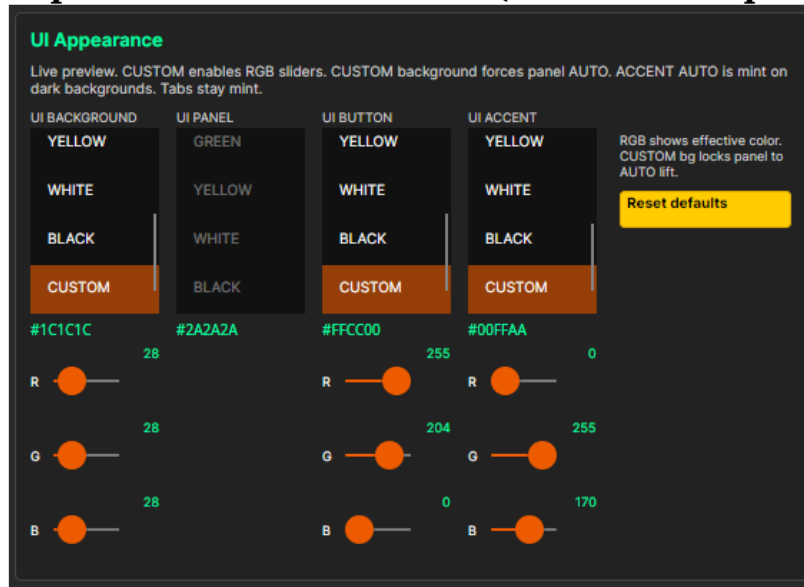
1. **RESET LOGS.** Clears the display.
2. **Pause.** Freezes the display; the log file keeps recording.
3. **CLOSE.**
4. **Log list.** Time-stamped messages from MSCC UI and the servers.

The log file is `~/ .local/share/MSCC-Avalonia/logs/mscc.log`. The servers keep their own logs in `~/ .local/mscc/`.

2.17 Choose Color (custom colors)

Any list on **SETTINGS** → **UI Appearance** (2.14), and the S/W **BACKGROUND** list (2.15), has a **CUSTOM** choice. Choosing it shows R, G and B sliders under that list; the change shows immediately.

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UI Appearance with CUSTOM chosen: R/G/B sliders under each list

1. **List** (UI BACKGROUND, UI PANEL, UI BUTTON, UI ACCENT). Pick **CUSTOM**.
2. **Color code** (for example #1C1C1C). The color actually in use.
3. **R** (0–255).
4. **G** (0–255).
5. **B** (0–255).
6. **Reset defaults**. Background BLACK, panel AUTO, button YELLOW, accent AUTO.

A CUSTOM background locks UI PANEL to AUTO (its list grays out). Tabs keep their mint color.

2.18 Keyboard and mouse tips

- MSCC UI has no menus and no keyboard shortcuts. A mouse with a wheel is strongly recommended, especially on laptops.
- Wheel over a VFO digit tunes that digit. Wheel over the VFO box (not a digit) tunes by Step. Double-click VFO A to change the Step. Wheel over a slider nudges it.
- Left-click the spectrum to tune.
- Hover over a control to see a short tip about it.



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□ **Appendix A: Updating the Radio Firmware (PSoC)**

The radio's main processor (a Cypress PSoC) holds two programs: the **application** (the radio firmware you update) and a small **bootloader** ("LOADER") that you normally never touch. Updating the application with a .cyacd file never changes the bootloader.

All radios must be updated to the latest firmware to use MSCC. After you connect, the title bar shows the firmware version (FW).

The firmware files. The optional **mscc-firmware** package (2.1) installs one dated .cyacd file (plus a .hex file) for each of the eight supported radios in /usr/share/mscc/firmware/<radio>/: Geminus-Legacy, Geminus-MKII, Proficio-Legacy, Proficio-MKII-ATU, Proficio-MKII-PTT, Ultimus-Legacy, Ultimus-MKII-ATU, Ultimus-MKII-PTT. For example Proficio-MKII-PTT/Proficio-MKII-PTT-20260822.cyacd. Use the .cyacd file. A newer mscc-firmware package brings newer files.

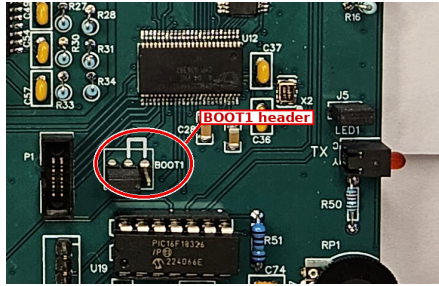
▲ Don't interrupt a firmware update. Once you click **Program**, don't unplug the USB cable, turn off the radio or close the program until the success message appears. It takes only a few seconds.

A.1 Updating the firmware over USB

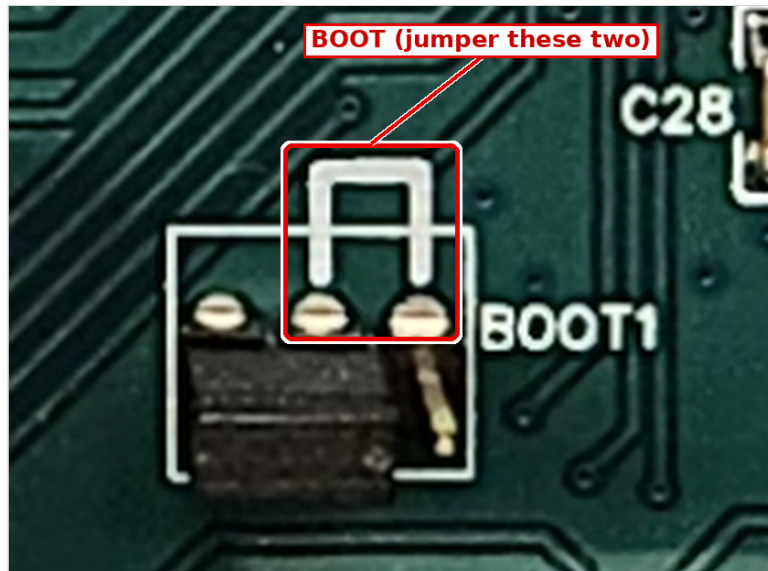
No extra driver is needed on Linux.

1. **Stop MSCC and the servers.** Disconnect (or close) MSCC UI. If the servers are still running, use **MSCC Stop**.
2. **Power the radio off.** Always move the shunt with the radio **off**. Leave the USB cable connected.
3. **Put the shunt on the boot pins.** Find the 3-pin **BOOT1** header. A small "U" printed on the silkscreen joins two of its pins: those are the boot pins. Put the plastic shunt (jumper) across them. The header's location differs by radio; the pictures show a Geminus MKII board.

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BOOT1 header on a Geminus MKII board



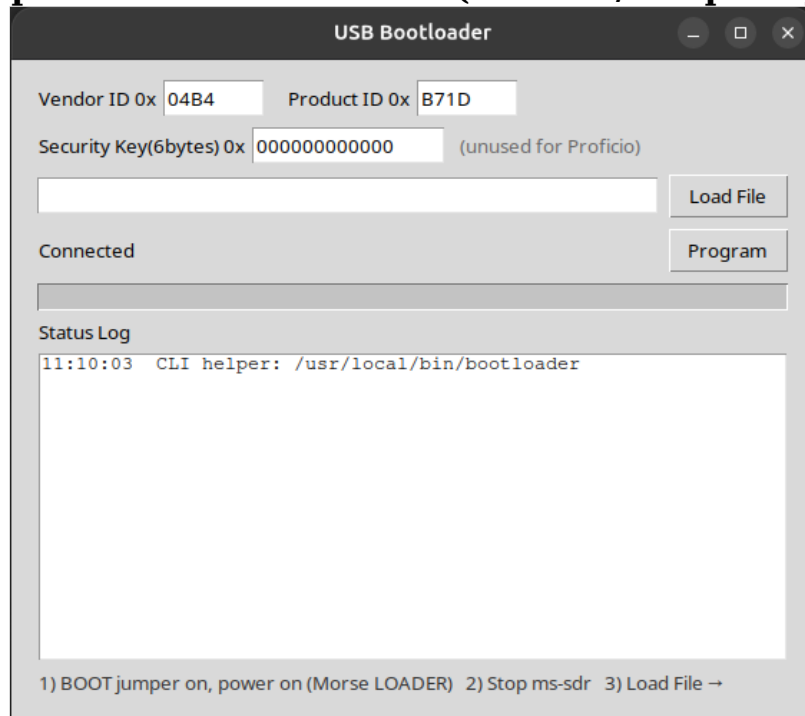
The printed U joins the two boot pins

Don't worry about the other pins. They are just grounds. A shunt on the wrong pins doesn't hurt anything.

4. **Power the radio on.** The LED should blink **LOADER** in Morse code. If it doesn't, power off and move the shunt to the two pins joined by the U.
5. **Start MSCC Firmware Upload** from the MSCC menu. Its window is titled **USB Bootloader**; the steps are printed along the bottom.
6. **Wait for Connected.** The line above the progress bar must show **Connected**.



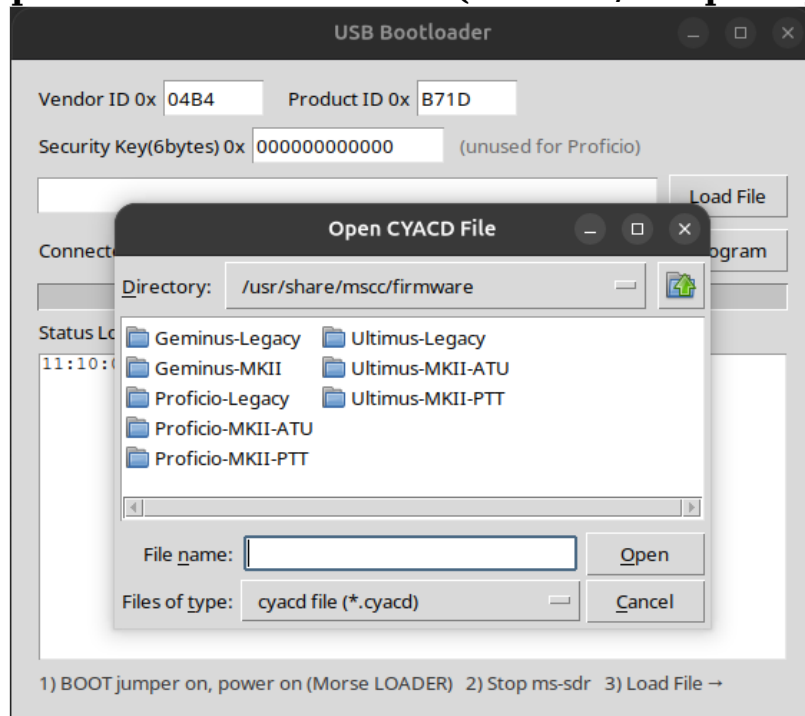
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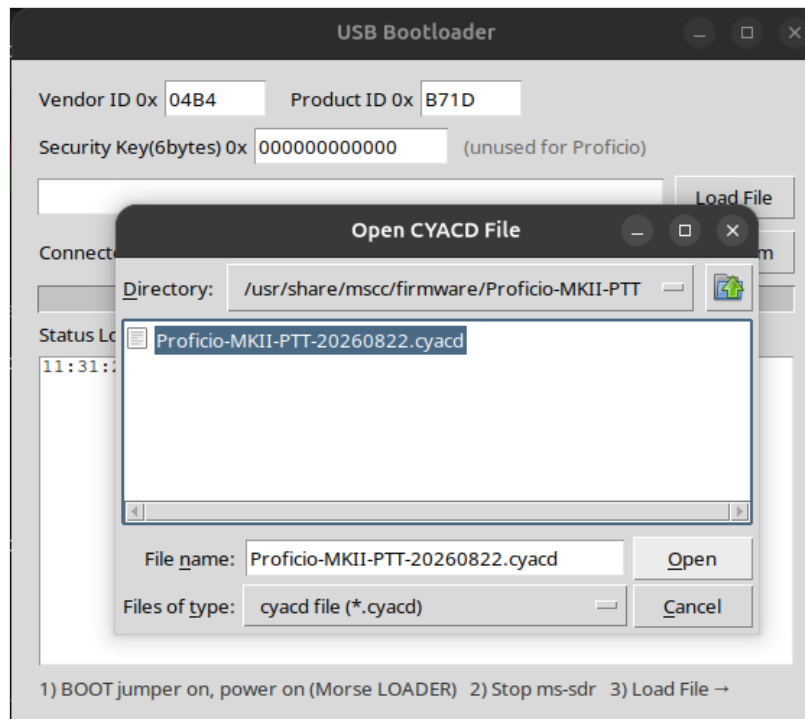
USB Bootloader, connected to the radio in boot mode

1. **Vendor ID** (04B4) and **Product ID** (B71D). Leave as they are.
 2. **Security Key**. Leave at all zeros (unused for Proficio).
 3. **File box** and **Load File**.
 4. **Connected / Disconnected**.
 5. **Program**.
 6. **Progress bar**.
 7. **Status Log**.
7. **Choose the firmware file.** Click **Load File**. The **Open CYACD File** window opens at /usr/share/mscc/firmware. Open your radio's folder, pick its .cyacd file and click **Open**. (If mscc-firmware isn't installed you get a warning and can browse to a file you downloaded.)

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Load File opens /usr/share/mscc/firmware: one folder per radio

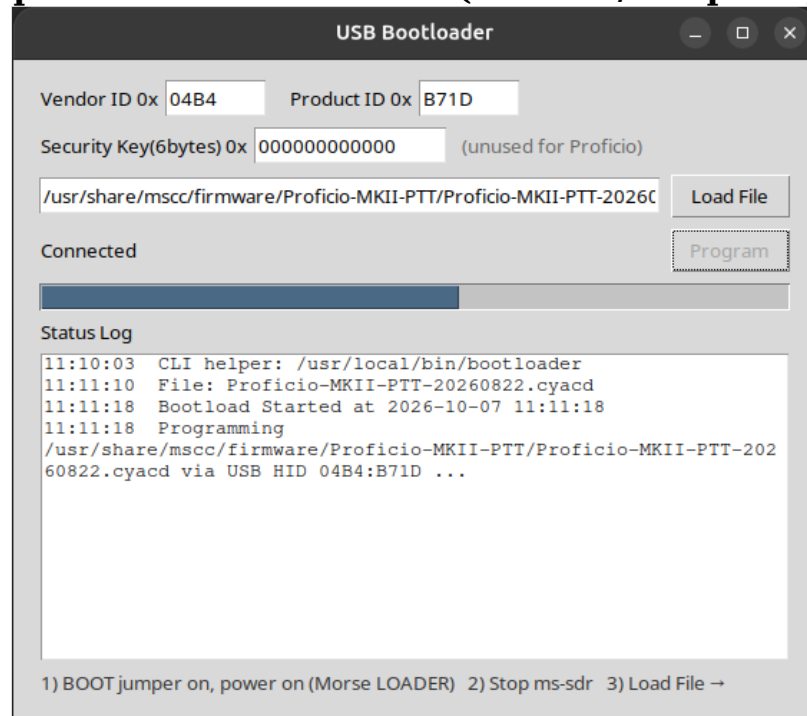


Pick the .cyacd file in your radio's folder

8. Click Program. The progress bar fills in a few seconds. **Don't interrupt it.**



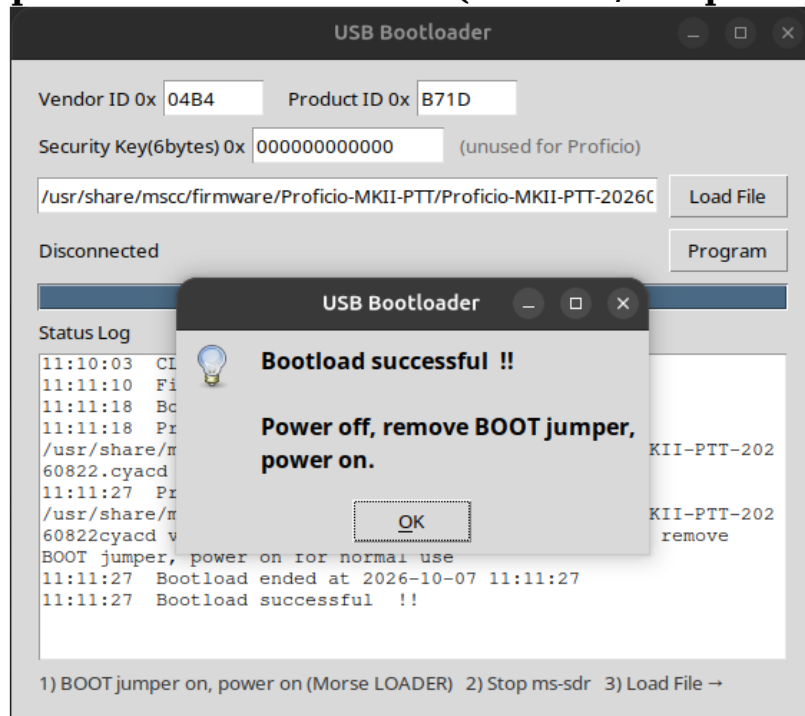
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Programming in progress

9. Check for the success message: **"Bootload successful !! Power off, remove BOOT jumper, power on."** Click OK.

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Bootload successful

10. Take the shunt off the boot pins with the radio powered **off**, then power on again.

11. Check the version. Start MSCC UI and connect; the title bar shows the new FW.

Terminal alternative for steps 5–9: `bootloader /usr/share/mscc/firmware/<radio>/<file>.cyacd` (radio in boot mode).



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□ **Appendix B: The Electronic Keyer**

MKII radios have a built-in electronic keyer. Plug your paddle into the **Key** jack.

- **Settings (CW tab, 2.6):** keyer mode (Straight, Iambic A or B), letter or element spacing, normal or reversed paddle, weight 25/50/75, speed 5–60 WPM (also on the right rail), pitch 400–1000 Hz, HOLD 1–500 ms.
- **CQ memories:** four slots, 48 characters each.
 1. Type your message, for example CQ CQ CQ DE W1ABC W1ABC K.
 2. Press **R** to store it in the keyer. The keyer keeps it even with power off.
 3. Press **P** to send it once.
 4. Touch the paddle to stop.
- **Farnsworth:** slower overall speed for memory playback; letters stay at SPEED. Paddle sending isn't affected.
- **External keyer or Legacy radio:** check **Use external electronic keyer (legacy)** on the CW tab and restart the servers. HOLD still sets the hang time.
- **POTENTIA / QSK:** amplifier PIN-diode T/R switching (2.6 item 10). Full QSK is not supported.



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□ **Appendix C: Installing and Upgrading**

- **First install:** Section 2.1 (Ubuntu) or Appendix G (Raspberry Pi), then MSCC Init (2.2).
- **Upgrading:** close MSCC UI (or Disconnect), then install the newer files with the same commands; you only need the packages that changed. You don't need to repeat `usermod` or log out again. If the servers were already running, run **MSCC Stop** then **MSCC Start** (or `mscc restart`).
- **What changed:** each kit has a `CHANGELOG.md`.
- **Your settings are kept:** MSCC UI settings in `~/.config/MSCC/`, server settings and calibration in `~/.local/mscc/`. Run MSCC Init again only if your audio devices or radio changed.
- **After a kernel update** the `/dev/tnt` ports may disappear; run `./install-mscc.sh` again so the virtual serial driver is rebuilt.



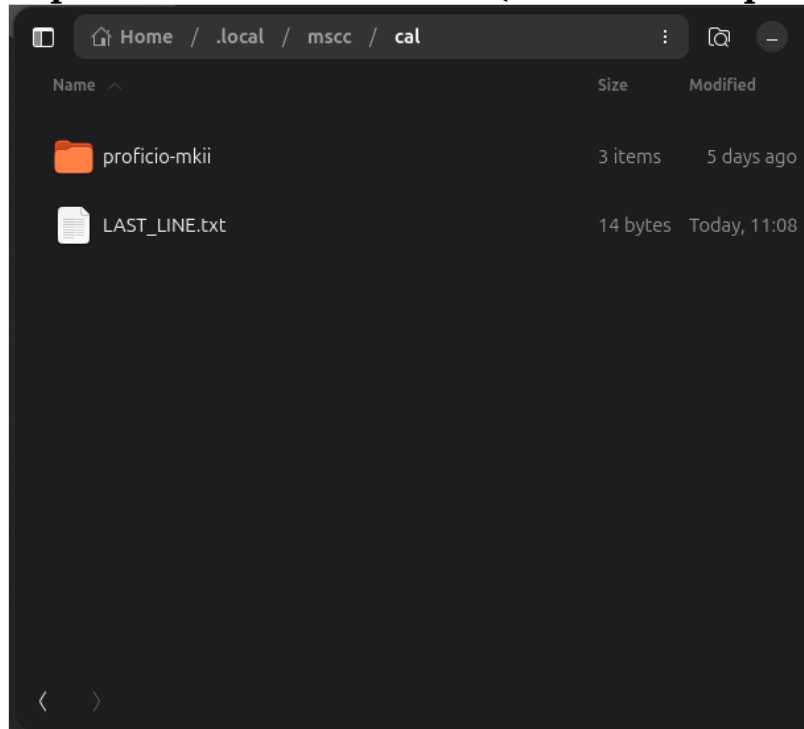
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□ Appendix D: Where Settings and Calibration Files Live

The ~ means your home folder (for example /home/stew). Folders starting with a dot are hidden; press **Ctrl+H** in Files to show them.

What	Where (Linux)
MSCC UI settings (window, display, last used, colors)	~/.config/MSCC/mscc-avalonia.ini
Favorites	~/.config/MSCC/mscc-favorites.ini
MSCC UI log	~/.local/share/MSCC-Avalonia/logs/mscc.log
Settings written by MSCC Init	~/.local/mscc/ (mscc.ini, comm-port.ini, operator-speaker.ini, ...)
Radio calibration (live)	~/.local/mscc/: iq.ini (TX IQ), recv-iq.ini (RX IQ), power_cal.ini (QRP), amplifier_cal.ini (AMP CAL), freq_cal.ini (frequency)
Per-radio calibration (Save settings)	~/.local/mscc/cal/ holds one folder per radio line (for example proficio-mkii) and LAST_LINE.txt. Save settings copies iq.ini, power_cal.ini and amplifier_cal.ini there.
Factory tables	/usr/share/mscc/factory/ (freq, iq, power), installed with the servers
Radio firmware files	/usr/share/mscc/firmware/<radio>/ (mscc-firmware package)
Server programs and logs	~/mscc/ and ~/.local/mscc/

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Files open on ~/.local/mscc/cal: radio-line folder and LAST_LINE.txt

Factory calibration by radio line. MSCC picks the factory tables from the firmware version's first number:

First number	Radio line
1	Proficio Legacy
2	Geminus MKII
3, 4	Proficio MKII (PTT and ATU share tables)
5	Geminus Legacy
6	Ultimus Legacy
7, 8	Ultimus MKII (ATU and PTT share tables)

- The servers detect which radio is connected. The same radio keeps its live calibration; a different radio gets its saved calibration from `cal/<radio line>/`, or the factory tables if none was saved.
- Factory tables are also used for **FREQ CAL RESET** and **TX IQ RESET ALL**.
- RX IQ (`recv-iq.ini`) and frequency calibration (`freq_cal.ini`) are not parked by Save settings.



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☐ Appendix E: Troubleshooting (FAQ)

“MSCC is not set up on this computer yet.” Run MSCC Init (2.2). You can click OK and keep going meanwhile.

Connect does nothing, or the spectrum stays empty. Check the radio's power and USB. Open **MSCC Status** and look for FAIL lines. With Launch unchecked, run MSCC Start first.

“WARNING: keep-alive lost”. Disconnect, check the radio's power and USB, and Connect again. If needed, `mscc restart`.

No firmware or core version in the title after connecting. Check USB and power, then Disconnect and Connect again (or restart the servers).

Launch or Remote is grayed out. Launch works only while disconnected and with Host 127.0.0.1. Remote is only for a radio on another computer.

Save settings says “Connect the radio first.” Connect and wait until the title shows FW, then press Save settings again.

No sound, or the wrong speaker or microphone. Run MSCC Init again (it stops the servers for you), pick the right devices in step 4, Write config, and start the servers. Check SETTINGS → Radio host.

WSJT-X can't open the serial port. Use the **CAT (WSJT-X)** port from SETTINGS (Appendix F), not the servers' port. Make sure you ran the `usermod` command and logged out and back in (2.1).

The /dev/tnt ports are gone after a kernel update. Run `./install-mscc.sh` again (Appendix C).

Firmware Upload: Load File warns that the firmware package is missing. Install `mscc-firmware_*_all.deb` (2.1), or browse to a `.cyacd` file you downloaded.

WSJT-X keys the radio but power is very low. Move the TX tone to about 1500 Hz; check AMP (off = normal QRP 5 W); keep ALC on (RX/TX tab); raise Digital Mic until ALC is just into the yellow.

CW paddle does nothing. With an external keyer or Legacy radio, check **Use external electronic keyer (legacy)** and restart the servers. For an MKII radio with a paddle in the **Key** jack, leave it unchecked.

Sending a problem report. Don't restart before collecting: a picture of MSCC Status, the MSCC UI log (`~/local/share/MSCC-Avalonia/logs/mscc.log`) and the server logs in `~/local/`



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mscc/. Then post the problem at <https://multusdr.groups.io/g/main/topics> and attach the logs as appropriate. All issues go there.



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□ Appendix F: WSJT-X Setup

WSJT-X talks to the MSCC servers through a virtual serial port; MSCC looks like a **Kenwood TS-2000**. MSCC UI doesn't have to be open for WSJT-X to work, but it is where you see the spectrum and change modes and power. This appendix stays at a high level.

Find your port first: SETTINGS → **Radio host (mscc-init)** → **CAT (WSJT-X)** (2.14). The orange hint line repeats the settings below.

WSJT-X setting	Value
Rig	Kenwood TS-2000
Serial port	The CAT (WSJT-X) port from SETTINGS, for example /dev/tnt0 (or its "also" name, for example /dev/ttyUSB10)
Baud / handshake	9600, none
PTT method	CAT
Mode	None . Then select DIG-U yourself in MSCC UI.
Split	None
Audio Input	VirtualB.monitor (Digital Input)
Audio Output	VirtualA (Digital Output)
TX tone	Around 1500 Hz

Steps

1. In MSCC UI, select **DIG-U**. The audio path switches to Digital.
2. Choose a Hi filter (1.4k or 1.0k are available in DIG-U).
3. **AMP:** off for barefoot 5 W; on only with your amplifier in line and AMP CAL done. **ALC:** keep it on.
4. In WSJT-X, click **Test CAT**, then **Test PTT**.
5. Press **Tune** in WSJT-X and adjust **Digital Mic** until the ALC meter is just into the yellow.

Only one program on each end of the CAT pair. Never point WSJT-X at the servers' end (CAT (ms-sdr)).

Other digital-mode programs: a more detailed Linux digital-modes appendix may be added in a later edition.



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□ Appendix G: Raspberry Pi Setup Differences

MSCC UI, MSCC Init and Firmware Upload look and work the same on a Raspberry Pi as on Ubuntu. Only the install kit differs.

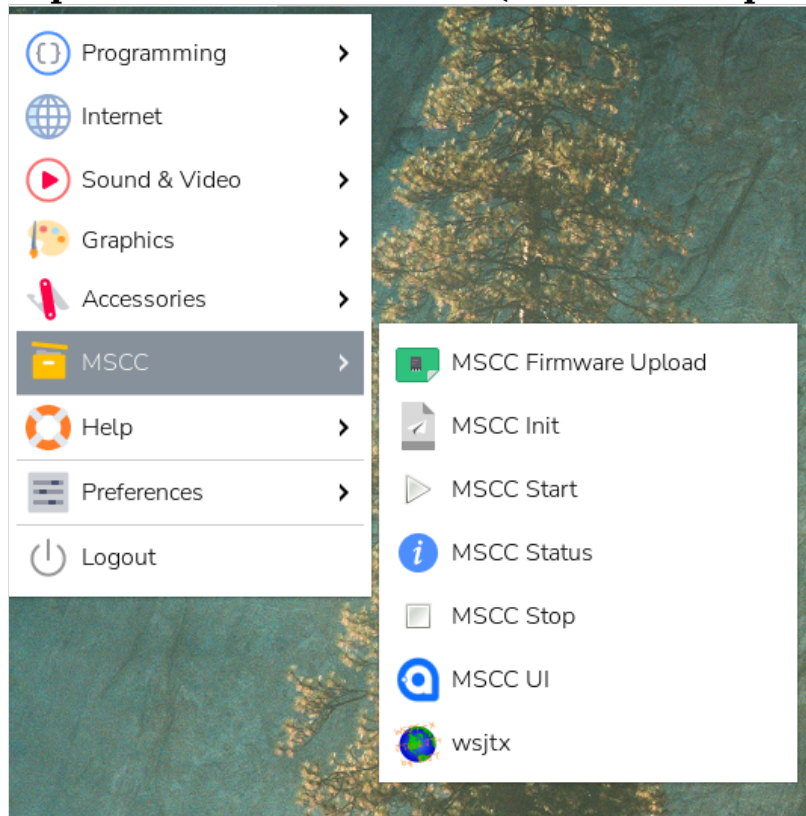
- **Computer:** Raspberry Pi 4 or 5 with 64-bit Raspberry Pi OS (desktop recommended).
- **Kit:** use the Raspberry Pi kit (`_arm64.deb` files). Never install `_amd64` files on a Pi.
- **Install order** (Terminal in the kit folder):

```
chmod +x install-mscc.sh
sudo apt install -y ./mscc-portaudio*_arm64.deb
./install-mscc.sh
sudo apt install -y ./mscc-init*_all.deb
sudo apt install -y ./mscc-ui*_arm64.deb
sudo apt install -y ./mscc-firmware*_all.deb
```

On the Pi, MSCC Init comes in the `mscc-init` package (GUI plus a text wizard) instead of `mscc-init-gui`. `mscc-ui` is optional if you operate the Pi from MSCC for Windows instead. The Pi kit may also contain `proficio-flash-tools`; it is optional and not needed for normal operation.

- **Then:** log out and back in, run **MSCC Init**, **MSCC Start**, and connect MSCC UI to `127.0.0.1:8888` (or MSCC for Windows to the Pi's address; see the Remote guide).
- **Menu:** the Raspberry start menu has an **MSCC** folder with the same entries as on Ubuntu (2.3).

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Raspberry Pi start menu, MSCC folder

- **Audio:** pick the Pi's audio device, USB headset or audio HAT in MSCC Init step 4.
- **CAT port:** the tnt pair may be the other way round on a Pi; always use the port SETTINGS shows.